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Indicators of Response to Exercise Training: A Systematic Review & Meta-Analysis

Ardavani A[1], Aziz H[1], Phillips BE[1,2,] Doleman B[3], Ramzan I[1], Mozaffar B[1], Atherton PJ[1,2], Idris I[1,2,4]

1. MRC-Versus Arthritis Centre for Musculoskeletal Ageing Research, University of Nottingham, Royal Derby Hospital Centre, Derby, UK.
2. NIHR, Nottingham BRC, University of Nottingham.
3. Department of Anaesthesia, University Hospitals Derby and Burton NHS Foundation Trust, Derby, UK.
4. Department of Endocrinology and Diabetes, University Hospitals Derby and Burton NHS Foundation Trust, Derby, UK.

Address for correspondence:
Dr Iskandar Idris
University of Nottingham Medical School,
Royal Derby Hospital,
Uttoxeter Road, Derby,
DE22 3DT, UK.
Email: iskandar.idris@nottingham.ac.uk

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Abbreviations

Maximal rate of oxygen consumption ($VO2_{max}$)

Endurance exercise training (END)

High intensity interval training (HIIT)

Resistance exercise training (RET)

Cardiorespiratory fitness (CRF)

Power output (PO)

World Health Organisation (WHO)

Quality of life (QoL)

Heart rate (HR)

Lactate threshold (LT)

Maximum heart rate (HR_{max})

Resting heart rate (HR_{rest})

Standard deviation (SD)

Type 2 diabetes mellitus (T2DM)

Risk of bias (RoB)

Odds ratio (OR)

Confidence interval (CI)

Inverse variance (IV)

Grading of Recommendations, Assessment, Development and Evaluations (GRADE)

Randomised control trial (RCT)

Body mass index (BMI)

Cross sectional area (CSA)

Smallest worthwhile change (SWC)

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Contributorship

Ardavani undertook data collection, analysis and wrote the first draft of the manuscript; Aziz undertook data collection and analysis; Phillips critically analysed data analysis, interpretation and provide technical advice on content; Doleman provided expertise in advanced meta-analysis technique; Mozaffar & Ramzan undertook bias assessment; Atherton: conceived the study and provided technical advice for content and writing; Idris: planned the study, provided overall supervision of the study conduct, analysis, interpretation and drafts. All authors contributed to the write up and approve the final draft.

Data sharing:

Additional data can be assessed by contacting the corresponding author.

Ethics:

Not applicable

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ABSTRACT

Background: Means-based analysis of maximal rate of oxygen consumption (VO_{2max}) has traditionally been used as the exercise response indicator to assess the efficacy of endurance (END), high intensity interval (HIIT) and resistance exercise training (RET) for improving cardiorespiratory fitness (CRF) and whole-body health. However, considerable heterogeneity exists in the inter-individual variability response to the same or different training modalities.

Objectives: We performed a systematic review and meta-analysis to investigate exercise response rates in the context of VO_{2max}:

i) in each training modality (END, HIIT and RET) versus controls, ii) in END versus either HIIT or RET, and iii) exercise response rates as measured by VO_{2max} versus other indicators of positive exercise response in each exercise modality.

Methods: Three databases (EMBASE, MEDLINE, CENTRAL) and additional sources were searched. Both individual response rate and population average data was incorporated through dichotomous and continuous data, respectively. A total of 29 studies were suitable for synthesis.

Results: END and HIIT both demonstrate a clear improvement in $VO_{2max,}$ with HIIT shown to elicit a non-significant greater improvement in VO_{2max} relative to END. Power output (PO) produces a greater response rate and effect than VO_{2max} in END and HIIT, with other exercise response indicators yielding variable and statistically non-significant results. Magnitude-based comparisons demonstrate a variation in exercise response indicator change per each training modality.

Conclusions: Heterogeneity of exercise response indicators, training modalities and use of dichotomous versus continuous datasets may explain variability in reported exercise responses. Exercise response indicators beyond VO_{2max} may feasibly be used in the future to assess for exercise response through a training modality-specific approach. Substantial inter-study heterogeneity in assessment protocols, exercise interventions and populations currently exists in the available literature. A deficiency in data concerning RET exists and constrains further interpretation of data concerning this training modality.

Keywords: Exercise response, Resistance Exercise, High Intensity Interval training, Endurance, VO2 max, Power output, meta-analysis

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Article summary:

Strength and limitations:

Strength:

- Robust analysis and synthesis of available evidence (randomised-control trial (RCT), case-control and cohort).
- We used the Mantel-Haenszel statistical method, the Random Effects model in anticipation of high intra-study heterogeneity.
- For continuous data, we utilised the inverse variance (IV) statistical method with the Random Effects model and established the outcome measure as Mean

Limitations:

- Significant heterogeneity of studies (exercise response indicators, training modalities, assessment protocols, population being studied and analytical methods) limits the ability to undertake more extensive meta-analysis of available data.
- Analysis had to be separated for continuous and categorical data, which further limits the number of studies for each analysis.

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INTRODUCTION

Physical activity in humans has been recognised to confer a beneficial effect on health since the time of Hippocrates and Galen[1,2]. The World Health Organisation (WHO) defines physical activity as "any bodily movement produced by skeletal muscles that requires energy expenditure"[3]. Physical activity has been shown to not only be cardio-protective[4], but prospective data has demonstrated an inverse correlation between increased physical activity and all-cause mortality[3,5,6]. Over the years, regular physical activity has been implicated in the prevention or management of a considerable number of chronic diseases, including cancer[7,8]. Physical activity has also been actively implemented as an intervention for age-associated frailty, resulting in a marked improvement in the quality of life (QoL) of older individuals[9-11], as well as improvements in the constellation of age-associated metabolic abnormalities which include dyslipidaemia, hyperglycaemia, hypertension and obesity[12-16].

Within the literature concerning structured exercise training as a form of physical activity, and away from specialist athletic training regimes, three broad variants of exercise training modalities are commonly described: i) Endurance exercise training (END), ii) high intensity interval training (HIIT), and iii) resistance exercise training (RET)[17,18]. Each of these training modalities is associated with a multitude of differing components, including content, volume, intensity, duration (training and recovery periods) and frequency[19-22]. As such, each modality is associated with distinct improvements in musculoskeletal, metabolic and/or cardio/vascular parameters[6-8]. Guidelines published by the UK Department of Health and Social Care in September 2019 advise that healthy adults should perform at least two instances of RET and a total of 150 minutes of moderate END on a weekly basis[23]. There is also some recognition of HIIT, as a reduced volume vigorous exercise can replace moderate END as long as this is accumulated in bouts of 10 minutes or more.

In terms of determining the efficacy of an exercise training programme, despite the aforementioned differences which will likely occur with different training regimes, an assessment of cardiorespiratory fitness (CRF) is often undertaken. In humans this is most commonly undertaken through assessment of the maximal rate of oxygen consumption (VO_{2max}), defined as the peak utilization of oxygen by metabolically active tissue [24,25]. Subsequently, the VO_{2max} plateau has become the gold standard for determining the maximal

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CRF of an individual, in addition to serving as a comparative marker of response following an exercise intervention[26-30].

Although it is well-established that improvements in multiple health parameters are generally observed in humans following a period of exercise training[27,31], the observance of individuals who do not demonstrate an improvement for a particular indicator above measurement error has emerged in multiple studies (often defined as "non-responders")[31,32]. Estimated to constitute up to ~20% of any given population for the primary expected physiological adaptation (e.g. hypertrophy for RET and VO_{2max} or insulin sensitivity for END)[32], numerous explanations have emerged to try and describe this phenomenon, ranging from innate factors to poor compliance[32,33]. However, simple explanations based on baseline characteristics or training compliance/intensity do not appear to be able to fully explain the marked heterogeneity in exercise adaptation [27]. Furthermore, a subset of individuals demonstrate a worsening of a given indicator below measurement error. with some describing these individuals as "adverse responders" [31,34]. If using this nomenclature, one study reported that an estimated 7% of people are described as being adverse responders in at least two parameters[33] , with commonly assessed variables (in addition to $VO_{2max)}$ including heart rate (HR), lactate threshold (LT) and power output (PO)[34,35].

It must however be acknowledged that the concept of non-responders, and certainly that of adverse responders, is not universally accepted with ongoing debate in the scientific community concerning the epistemological validity of this concept[27,31]. Some of this debate is centered on the definitions of these concepts, with the work of Mitchell *et al.* 2014 claiming there are no non-responders to RET as all individuals demonstrated at least one positive adaptive response of the many that were measured [75]. This definition of a non-responder is however strikingly different to that used by Phillips and colleagues in their work looking at molecular networks of exercise adaptation. In this study, non-responders were classed as those who did not demonstrate significant hypertrophy during 20-weeks RET, but may have displayed other improvements such as strength or vascular conductance. As such, a number of these individuals would have been non-responders in one study but not in another. This apparent uncertainty has been further exacerbated by the observation of a dissociation between VO_{2max} and other exercise response indicators (including blood lactate and

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maximum heart rate (HR_{max})) with END[36]. Furthermore, evidence of a disparity in indicator-based responses has been demonstrated in analyses of outcomes following RET exercise[37] with hypertrophic gains not necessarily representing changes in muscle function, for example.

Of the three aforementioned exercise modalities and with respect to improvements in CRF, END and HIIT are each recognised as having a significant and positive effect overall [18, 64]. In contrast, the benefits of RET are distinct from this and are traditionally considered to be improvements in strength and skeletal muscle hypertrophy[18]. Although early evidence determined that RET did not confer any improvement in VO_{2max}[73], a more recent comprehensive assessment through a narrative systematic review comprising 17 studies, concluded that improvements in VO_{2max} may be observed with RET in previously untrained individuals irrespective of age[74]. As such, sedentary individuals would conceivably benefit from a concurrent improvement in both CRF and the acknowledged skeletal muscle-based improvements associated with RET[74].

As a result of the current paradigm within the literature, there exists an uncertainty concerning the suitability of different interventions to elicit an exercise response. In addition to the academic consideration of what may constitute as a "response" to any particular intervention, a determination of whether alternative markers of response to VO_{2max} exist in each of the three modalities has not yet been undertaken through a systematic approach.

Therefore, through a combined systematic review and meta-analysis strategy, this study appraises the available evidence of studies in order to answer the following research questions:

1. In untrained human adults, is each exercise modality (END, HIIT and RET) more effective than controls in eliciting an improvement in CRF based on VO_{2max}?
2. In untrained human adults, is END more effective than either HIIT and RET in eliciting an improvement in CRF based on VO_{2max}?
3. In untrained human adults and per each exercise modality, do other measures of exercise response (HR_{rest}, HR_{max}, LT and PO) elicit a similar rate of exercise response when compared with VO_{2max}?

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METHOD

The population assessed in this study was human adults between the ages of 18 to 80 years. As the extent of exercise response is noted to be similar between both males and females[26], both sexes were included in all analyses. The investigated intervention training modalities were END, HIIT and RET. To permit the investigation of the three research questions, the comparisons and outcome measures used were i) intervention (END, HIIT or RET) versus control using VO_{2max} as the response indicator, ii) END versus HIIT or RET using VO_{2max} as the response indicator, and iii) VO_{2max} versus HR_{rest}, HR_{max}, LT or PO in each of END, HIIT and RET. In anticipation of a paucity in data, the outcome measures included in this study were divided into categorical (dichotomous) data based on the study-specific definition for exercise response threshold (Domain A) and continuous data represented by mean±standard deviation (SD) values (Domain B).

The inclusion criteria are all published clinical trials (randomised-control trial (RCT), case-control and cohort) in previously untrained healthy or obese individuals or individuals with type two diabetes mellitus (T2DM) without physical impediment, and the utilisation of VO_{2max} as an endpoint. Additionally, to facilitate a comparison for the Domain A and Domain B data types, eligible studies must contain a study-specific division between "response" and "non-response" or contain a comparison group, respectively. The exclusion criteria are limited to qualitative studies, non-human studies, studies where participants are younger than 18 or older than 80 years of age, physically impaired individuals, pregnant volunteers, or participants with any established cardiovascular, renal, musculoskeletal, neurological, malignant or pulmonary disease.

Three electronic online literature databases (EMBASE, MEDLINE, CENTRAL) were utilised as primary data sources. Search strategies for EMBASE and MEDLINE were undertaken (Supplement 1). The search strategy for CENTRAL was "VO2 exercise response". To screen for clinical trials not captured in the above search strategies, and to definitively address the anticipated deficiency in RET intervention studies, two further search procedures were performed through PubMed. The search terms in each were "VO2 exercise response" and "resistance exercise response variability". Only clinical trials were selected. Furthermore, grey area sources were sought in addition to the above. The search phrase "exercise response

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VO2" was submitted to the Google search engine (147 results, seven relevant studies). An additional search was made through ClinicalTrials.Gov (completed studies, adult, all sex, "exercise response" search string, only studies with preliminary or final results selected, 1308 results, six relevant studies). A further search through the reference lists of studies selected for synthesis was implemented to further address any undetected primary sources.

The MeSH terms undertaken in this study are in Supplement 1. Two of the primary database sources (EMBASE, MEDLINE) were searched by two independent researchers (AA, HA), with CENTRAL, PubMed, Google, ClinicalTrials.Gov and reference list review searches and further selection stages being undertaken by one researcher (AA).

In the first screening procedure, all studies were initially assessed on their implementation of END, HIIT and/or RET as a primary intervention and their utilisation of at least one response indicator, where VO_{2max} was mandatory. In accordance with the findings of a meta-analysis published by Bacon *et al.* 2013 that determined VO_{2max} trainability is increased with prolonged intervention periods[38], studies with a duration of less than three weeks were excluded from the analysis. Following the completion and compilation of the studies obtained from the preliminary data source searches, duplicates were highlighted and the novel studies were spliced into a new list. Thereafter, any new studies obtained from the other search strategies were added to a continuously updated version of this list. The extracted summary data for all studies that were deemed suitable for synthesis included the study title, primary author, year of publication, association with any other studies or trials, study design type, target population characterisation, assessed interventions, duration of intervention, intervention detail (including means of exercise, frequency, volume parameters and/or intensity), primary and secondary endpoints, criteria for exercise response (if applicable), data type (categorical versus continuous) and suitability for any combination of the intended analyses specific for each research question (I to III).

The data specific to response rate included exercise modality (END, HIIT and/or RET), the parameters utilised (VO_{2max}, HR_{rest}, HR_{max}, LT and/or PO) and the data type (categorical versus continuous data). With the categorical data (Domain A), the response and non-response rate for a given parameter was entered as indicated in the original study and no *post hoc*

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thresholds were applied. With the continuous data (Domain B), only the post-intervention mean and SD for intervention and control (or other intervention) groups were entered for comparison. The information per study was separately recorded in three tables, each pertaining to one of the three research questions established.

The Cochrane Risk of Bias (RoB) 2 template was implemented for randomised studies [39]. Qualitative assessment of the studies was undertaken by three researchers. Both study and outcome level outputs were produced. The resulting outputs were generated through Review Manager (RevMan) v5.3.5. Additionally, case-control studies were assessed using the CLARITY McMaster University Risk of Bias assessment framework. These were independently undertaken by three researchers (AA, BM, IR).

Two principal summary measures are produced in this meta-analysis. For Domain A (categorical data), we calculated the odds ratio (OR) from the study-specific response versus non-response count. Analysis method settings included the Mantel-Haenszel statistical method, the Random Effects model in anticipation of high intra-study heterogeneity and a 95% confidence interval (CI). The resulting data is reported here with OR, 95% CI, effect size and p-value data for both individual studies and pooled data. For Domain B (continuous data), the inverse variance (IV) statistical method was selected with the Random Effects model and established the outcome measure as Mean Difference and a 95% CI based on post-intervention mean±SD data in the assessed groups. All data was reported with IV values and 95% for all individual studies, with pooled data reported with the addition of Z data and p-values.

Forest plots were generated for all datasets which contained at least two data-points. All statistics and forest plots were produced with RevMan v.5.3.5. Data per study was manually entered into each of the variable listed tables outlined earlier. No *post hoc* data merging between studies was undertaken. Measures of statistical heterogeneity were calculated using the I^2 statistic through RevMan v.5.3.5 and are reported within the produced forest plots. Certain studies permitted the inclusion of multiple groups separately for comparative purposes (Table 1).

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In order to assess for publication bias, the Grading of Recommendations, Assessment, Development and Evaluations (GRADE) approach[40] was adopted on a study-specific level and was assessed by two independent researchers (AA, IR). Due to the limited number of studies per area of analysis, statistical assessment of publication bias, meta-regression and Trial Sequential Analysis was infeasible.

Patient and public involvement.

Patients and or public were not involved in undertaking and devising this systematic review and meta-analysis.

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RESULTS

A total of 3,268 studies were generated from the identification stage. A total of 29 studies - 20 RCTs and 9 case-control studies - were deemed suitable for inclusion in our qualitative and quantitative appraisal (Figure 1). The publication dates ranged from 1991 to 2018, where the majority of the assessed studies were not associated with any other trials. A total sample size of 1,937 individuals was assessed.

The study characteristic data for all studies are presented in (Table 1). The assessed populations were found to be heterogeneous with respect to biological sex, degree of physical fitness, body mass index (BMI) and age. The majority of studies featured a comparison between END and HIIT and included either END or HIIT within their analysis, with only three studies assessing the response rates for RET. One study[44] was found to employ a parallel assessment of exercise intensity and volume. The implemented exercise protocols exhibited substantial heterogeneity, with only two studies[42,43] displaying a congruent basis for their assessment due to their common derivation from the DREW study. The intervention period varied between three to 52 weeks. In accordance with our inclusion criteria, the majority of studies utilised VO_{2max} as a primary endpoint, with most studies also incorporating data pertaining to body composition. 12 of the 29 studies defined exercise responsiveness through differing thresholds of VO_{2max} change, where three of these 12 defined any improvement from VO_{2max} baseline ($\Delta > 0\%$) as evidence of a positive adaptation in their cohorts. [42,43,46]

Of the 29 studies, 11 were found to contain “absolute data” (defined herein as dichotomous data on the basis of the study-specific exercise response criteria) and 21 contained mean±SD data for each of the assessed groups. The majority of the studies contained sufficient data for inclusion in the assessment of at least one of the three research questions, with those studies containing “absolute data” categorised as “Domain A” and dichotomous data used for “Domain B” assessment. The RoB summaries are enclosed in Supplements 2 to 4, which demonstrate varied degrees of bias across the assessed domains. Furthermore, GRADE score appraisal of each study was undertaken (Supplement 5). 81 data-points were obtained from these studies for inclusion in the appropriate analyses.plots (Figures 2-5).

Exercise Responsiveness versus Controls Using VO_{2max} (Analysis 1)

END vs. CON

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The majority of data-points (10 of 11) demonstrated a favouring of END over controls, with only one from a single study[49] yielding the opposite outcome (Figure 2). Statistical significance was achieved in three of the four 'Domain A' data-points, although these were all derived from the same study[43] (Figure 2A). Similarly, these studies demonstrated a similar effect size, which was not replicated in another study[30] although favouring of the intervention was still observed (OR = 0.88). The confidence intervals across data-points do not overlap and statistical significance is not achieved overall (p = 0.06). Both studies utilised differing definitions for non-responsiveness (VO_{2max} Δ > 0% and 5% for Earnest *et al.* 2011 and Pandey *et al.* 2015, respectively). As such, 'Domain A' data-points for this element of Analysis 1 yielded inconclusive results.

In contrast, the 'Domain B' data-points exhibit partial CI overlap and overall statistical significance (p < 0.0001), with a positive effect size and a narrow CI (IV = 5.59, 95% CI = 3.84 - 7.35) (Figure 2B). Therefore, in contradistinction with the outcome observed in 'Domain A' data, the analysis with 'Domain B' data demonstrates an unequivocal improvement in VO_{2max} through END as an intervention.

HIIT vs. CON

Only one study [29] was available for assessment in 'Domain A', yielding a single data-point, which demonstrated a statistically significant (p = 0.03) positive overall effect through the intervention (OR = 0.13, 95% CI = 0.02 - 0.78). Eight studies contributing nine data-points were present in 'Domain B' (Figure 2C). The majority (seven of nine) demonstrated a favouring of the intervention (Weatherwax *et al.* 2018's 'Individualised' and Metcalfe *et al.* 2012's data-points revealed a control-favouring and indeterminate overall outcome, respectively). Partial overlap in CI was observed across studies. The overall effect size was significantly positive with a narrow CI (IV = 3.07, 95% CI = 0.61 - 5.53; p = 0.01). Summarised, both the 'Domain A' and 'Domain B' data demonstrate an improvement in VO_{2max} using HIIT versus control groups.

RET vs. CON

A single study [30] was present in 'Domain A', which yielded a no significant difference (OR = 0.79, 95% CI = 0.35 - 1.82, p = 0.59). Similarly, one study[18] was found to yield an outcome which favoured control over intervention (IV= -1, 95% CI= -3.69 - 1.69). As such, the available

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data concerning the cardiopulmonary improvements afforded by RET are equivocal at present.

END versus HIIT and RET Responsiveness Using VO_{2max} (Analysis 2)

END vs. HIIT

Two studies, each contributing a single data-point, were present in 'Domain A' (Figure 3A). Individual effect sizes were contradictory (Ross *et al.* 2015 OR = 14.08 favouring END, Bonafiglia *et al.* 2016 OR = 0.68 favouring HIIT), and no significant difference was seen (95% CI = 0.10 - 56.88; p = 0.59). 'Domain B' contained 12 data-points from an identically-numbered set of studies (Figure 3B). Similar to the 'Domain A' analysis, the overall effect size was small and non-significant between END and HIIT (IV = -0.24, 95% CI = -1.75 - 1.27; p = 0.76).

END vs. RET

Two studies, each contributing a single data-point, were present in 'Domain A' (Figure 3C). The individual effect sizes were contradictory (Hautala *et al.* 2006 OR = 0.27 favouring END, Pandey *et al.* 2015 OR = 1.11 favouring RET). Minimal overlap between CI's was observed, with an overall effect to be non-significant (OR = 0.53, 95% CI = 0.14 - 1.99; p = 0.35). Only one study was present within 'Domain B'[18]. The outcome favoured END (IV = 4.3, 95% CI = 1.73 - 6.87).

Exercise Responsiveness Using Other Exercise Response Indicators (Analysis 3)

END & HR_{rest}

No data was available under 'Domain A'. 'Domain B' featured two studies [26,18] . Neither of these studies demonstrated a statistically significant result (Figure 4A).

END & HR_{max}

Two studies were available for 'Domain A'[47,27] (Figure 4B). The individual OR are contradictory (OR = 1.54, 95% CI 0.28 - 8.18 and OR = 0.56, 95% CI = 0.16 - 1.91; p=0.65). Four studies contained appropriate data for 'Domain B', with one exclusion[18] due to a fixed assessment speed (Figure 4C). Predominant CI overlap is observed and the overall effect was found to be not statistically significant (IV = -0.16, 95% CI= -3.01 - 2.7; p=0.91).

END & LT

Two studies were available for 'Domain A'[47,27] (Figure 4D). Partial CI overlap is observed, with both demonstrating similar directionality in favouring LT over VO_{2max} (OR = 4.31, 95% CI = 0.42

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- 43.73, OR = 1.48, 95% CI= 0.43 - 5.05 respectively). The overall effect size was OR = 1.87 (95% CI= 0.63 - 5.53; p = 0.26). Only one study[60] contained data in 'Domain B', where END was slightly favoured over control (IV = 0.6, 95% CI = -0.28 - 1.48).

END & PO

Five data-points were present within 'Domain A', however, these were all derived from the same study and differed only with respect to training volume [27] (Figure 4E). Substantial CI overlap observed. A significant effect size (OR = 1.97, 95% CI = 0.87 - 4.48; p = 0.10) was observed. Two data-points were present in 'Domain B', each being derived from different studies[59,60] (Figure 4F). Although substantial overlap in CI is observed, the range associated with the overall effect size is wide (IV= 35.49, 95% CI= - 4.23 - 75.20; p = 0.08).

HIIT & HR_{rest}

No 'Domain A' data was available for assessment. Within 'Domain B', three data-points from two studies[18,41] were available for analysis (Figure 5A). CI overlap is predominant, with an overall a non-significant effect which favours HIIT over controls (IV = - 4.6, 95% CI = -10.63 - 1.43, p = 0.14).

HIIT & HR_{max}

A single study[27] was present in 'Domain A', where VO_{2max} demonstrated a non-significant response rate compared with HR_{max} (OR = 0.53, 95% CI = 0.15 - 1.92; p = 0.34). Four studies were suitable for analysis in 'Domain B', where one[18] was omitted due to the measurement of submaximal HR using a fixed training velocity [18,29,41,60] (Figure 5B). Significant overlap in CI is observed, with an overall effect size that showed a non-significant difference between controls and HIIT (IV = -1.3, 95% CI = -6.03 - 3.43, p = 0.59).

HIIT & LT

Three studies[27,46,45] were available in 'Domain A' (Figure 5C). Near-complete overlap in CI is observed. Further, VO_{2max} demonstrated a non-significant response rate compared with LT (OR = 0.7, 95% CI = 0.34 - 1.42; p = 0.32). Only one study[60] was available in 'Domain B', with HIIT demonstrating a higher LT in comparison to controls (IV = 0.5, 95% CI = -0.21 – 1.21).

HIIT & PO

Only one study [46] was available in 'Domain A' , where PO yielded a significant higher exercise response rate in comparison with VO_{2max} (OR = 4.38, 95% CI = 1.28 - 14.97; p = 0.02). Four

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studies [29,52,59,60] were present in 'Domain B', demonstrating a wide inter-study CI, and an effect size which showed a non-significant trend favouring HIIT over no intervention (IV = 39.23, 95% CI= -3.07 - 81.54; p = 0.07) (Figure 5D).

RET

Unfortunately, no data concerning the difference in responsiveness rates between VO_{2max} and other indicators were present in 'Domain A'. Only two data-points from two studies[18,54] were present in 'Domain B', where HR_{rest} was found to result in a decrease in RET when compared to controls (IV = -5, 95% CI= -7.25 - -2.75) and HR_{max} increased in RET relative to controls (IV = 4.1, 95% CI = -4.37 - 12.57).

Exercise Indicator Rank-Order Comparison

Based on the results from 'Domain B' data, each of the five assessed indicators (VO_{2max}, HR_{rest}, HR_{max}, PO, LT) exhibit differing patterns of change in each exercise modality when compared with controls (Table 2). Unfortunately, as all of the outcomes listed herein failed to meet statistical significance, these results are equivocal.

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Discussion

This systematic review and meta-analysis investigated the variability in reported responses to END, HIIT and RET. We found that various factors such as training modality, volume and intensity, response indicators and use of dichotomous versus continuous datasets may affect the reported exercise training responses. Overall, both dichotomous and continuous datasets for HIIT demonstrated a significant effect size using VO_{2max} versus controls. However, variability in reported responses to END and RET (versus controls) was observed, dependent on the use of dichotomous versus continuous datasets. Overall, END resulted in a significant improvement in CRF versus controls using $VO_{2max,}$ with END slightly outperforming HIIT in terms of effect size. This finding contradicts an earlier meta-analysis performed in 2015, which assessed the responsiveness of END and HIIT in healthy adults between the ages of 18 to 45 years[64]. This difference may be due to our inclusion of studies including older cohorts, potentially reflecting an age-determined effect differential between exercise modalities. Likely a consequence of the historically-attested reduced effect afforded by RET in improving CRF, a deficiency in data concerning RET and exercise response indicators of CRF was observed which limits our ability to form comparisons with END and HIIT data. As a consequence of this, the currently-available data cannot elucidate the potential for an improvement in $VO2_{max}$ in untrained individuals through RET[74].

A comparison of exercise response rates between END with HIIT or RET using VO_{2max} through dichotomous data was inconclusive due to the significant heterogeneity observed in the studies included for synthesis.

A statistically significant outcome was however reported in a study using dichotomous data which showed a higher response rate of PO versus VO_{2max} in HIIT. Conversely, the analysis of continuous data demonstrated an overall non-significant improvement in PO with HIIT or END in comparison to control. This finding is partially in accordance with a recent cohort study in eleven moderately-trained cyclists which determined that PO exhibited a more pronounced relationship with athletic performance than VO_{2max}[65]. Our inconclusive outcome concerning LT appears to contradict earlier work, where a moderate positive correlation ($r^2 = 0.39$, $p < 0.05$) was inferred from a prior cohort study assessing the effect of END in sedentary males[66]. No data was assessed concerning the relationship between PO and RET, although this will

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ostensibly demonstrate an improvement due to the recognised development of skeletal muscle through type II fibre cross-sectional area (CSA) increases, sarcoplasmic hypertrophy and neuromuscular efficiency [76].

Heterogeneity in study design, intervention(s) and population characteristics were a recurrent feature in the assessed literature. A paucity in congruent measures restricted our ability to perform the appropriate multi-study sub-analysis. Similarly, the majority of comparisons incorporated data from studies that implemented differing exercise training protocols. This heterogeneity is reflected in the statistical data, where statistical heterogeneity is demonstrated (I^2 > 50%) in all Analysis 1 and 2 forest plots, in addition to three of ten in Analysis 3. Furthermore, exercise response is a complex trait, with multiple innate and environmental factors implicated[67]. As such, the absence of further participant differentiation due to the paucity in data, in combination with an enumeration of the known variables in exercise responsiveness ("trainability"), represents a source of confounding[68].

The statistical outcome measure in the majority of studies was based on either dichotomous or continuous data. Although our parallel assessment of both data types sought to maximise the possible inferences from currently available data in the literature, a larger degree of concordance in data types would have facilitated a larger sample size and potentially permitted inter-study sub-analysis. Generally, there was a paucity in data with respect to dichotomous data (assessment of exercise response using a pre-determined threshold).

Differing criteria for exercise non-responsiveness in studies utilising VO_{2max} using dichotomous data was also observed, resulting in added variability, and likely responsible for some of the observed disparities. This limitation was also present with other indicators. Lastly, cardiovascular-dependent indicators of exercise response typically require longer to elicit a change in comparison to VO_{2max}[69]. As such, we speculate that HR_{rest} and HR_{max} may demonstrate different results if only longer intervention periods were analysed.

The lack of consistency with respect to the definition of exercise non-response[31] using VO_{2max} requires addressing[70]. The utilisation of Δ > 0%, which was implemented in three studies, in our opinion is inappropriate given the acknowledged issue of measurement error[70]. Bonafiglia *et al.* 2018 derived an alternative approach, using response CI and the smallest worthwhile change (SWC)[71]. As described previously, the majority of the studies included for synthesis

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(26 of 29) presented their findings in either a dichotomous or continuous format. Producing both would simultaneously broaden the scope of future analyses, or provide researchers with additional latitude in data selection. Further elucidation on the relationship between VO_{2max} and PO across the different exercise modalities in the general population may indicate differing patterns of responsiveness. Similarly, the relationship between extrinsic factors and exercise indicators may yield differing effects on chronic adaptation in each exercise modality. Additionally, the relationship between the role of exercise indicators beyond VO_{2max} and the interference effect (the frequently-observed diminishment of RET-specific adaptations to muscle size and function in a concurrent training setting)[72] may reveal novel or anticipatory patterns, which may predict this outcome.

In conclusion, END and HIIT elicit an observable improvement in VO_{2max} versus controls. The extent of exercise non-response between and across the different modalities using VO_{2max} could not be ascertained. Similarly, comparisons between most alternate indicators of CRF yielded equivocal results. However, our study has demonstrated an apparent role for PO as a potential alternative indicator to VO_{2max} in exercise training scenarios.

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Figure legends:

Figure 1

Study screen, eligibility and inclusion flowchart

Figure 2

Analysis 1 forest plots depicting the exercise responsivness using VO_{2max} in END versus controls ('Domain A'/dichotomous data) (A), using VO_{2max} in END versus controls ('Domain B'/continuous data) (B), using VO_{2max} in HIIT versus controls ('Domain B'/continuous data) (C)

Figure 3

Analysis 2 forest plots depicting the exercise responsivness using VO_{2max} in END versus HIIT ('Domain A'/dichotomous data) (A), using VO_{2max} in END versus HIIT ('Domain B'/continuous data) (B), using VO_{2max} in END versus RET ('Domain A'/dichotomous data) (C)

Figure 4

Analysis 3 forest plots depicting the exercise responsivness using HR_{rest} in END versus control ('Domain B'/continuous data) (A), using VO_{2max} in END compared to HR_{max} ('Domain A'/dichotomous data) (B), using HR_{max} in END versus control ('Domain B'/continuous data) (C), using VO_{2max} in END compared to LT ('Domain A'/dichotomous data) (D), using VO_{2max} in END compared to PO ('Domain A'/dichotomous data) (E), using PO in END versus control ('Domain B'/continuous data) (F)

Figure 5

Analysis 3 forest plots depicting the exercise responsivness using HR_{rest} in HIIT versus control ('Domain B'/continuous data) (A), using HR_{max} in HIIT versus control ('Domain B'/continuous data) (B), using VO_{2max} in HIIT compared to LT ('Domain A'/dichotomous data) (C), using PO in HIIT versus control ('Domain B'/continuous data) (D)

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STUDY	YEAR	DESIGN	N	CHARACTERISTICS	INTERVENTION	Exercise Protocol (Modality, Intensity, Volume, Frequency)	Intervention Period	Comparison Type	Primary Endpoint	Secondary Endpoint	RESPONSE CRITERIA
Weatherwax [41]	2018	RCT	39	Male and female. Less than 30 minutes activity in 3 days per week. 30 to 75 years old.	HIIT	Stationary bike, elliptical machine or treadmill. Stepwise progression in intensity depending on exercise group.	12 weeks	Standardised vs. Individualised	VO_{2max}	Physical activity, sitting time, Hrrest, Hrmax, body measurement parameters, dietary intake	VO2max Δ > ME (4.7%)
Sisson [42]	2009	RCT	310	Female only (post-menopausal). Sedentary. 45-75 years old. multiple ethnicities.	END	Recumbent cycle ergonometer or treadmill (alternating); 50% baseline VO2max, 3-4 sessions/week	24 weeks	4 vs. 8 vs. 12kcal/kg/week	VO_{2max}	N/A	VO2max Δ > 0%
Earnest [43]	2011	RCT	251	Female only (post-menopausal). Sedentary. 45-75 years old. Caucasian only.	END	Recumbent cycle ergonometer or treadmill (alternating); 50% baseline VO2max, 3-4 sessions/week	24 weeks	control vs. 4 vs. 8 vs. 12kcal/kg/week	VO_{2max}	N/A	VO2max Δ > 0%
Pandey [30]	2015	RCT	202	Male and female. T2DM. 30-75 years old.	END	Treadmill; 50-80% VO_{2max}, 3 session/week	36 weeks	control vs. LISS vs. RET vs. both	VO_{2max}	HbA1c, body measurement parameters, S&DBPrest SBPeak, Hrrest, RER, insulin use	VO2max = Δ > 5%
					RET	Full body training, compound movements (upper, lower, abdominal); 10-					

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						12 repetition range, 3 sessions/week					
Montero [28]	2017	RCT	78	Male only. Sedentary. Healthy. 18-35 years old.	END, HIIT, mixed	Recumbent cycle ergonometer; average of 65% of Wmax (between four intensity profiles), range from 1-5 sessions/week	6 weeks	Exercise frequency (1 vs. 2 vs. 3 vs. 4 vs. 5 sessions/week)	VO_{2max}, W_{max}	haemoglobin mass, plasma volume, red blood cell volume, blood volume, body measurement parameters, mitochondrial volume density	VO2max & Wmax = >1 T.E.
Bonafiglia [27]	2016	RCT (crossover)	21	Male and female. Recreationally active only. Age range not stated.	END	30 minutes cycling at 65% of VO2max, 4 sessions/week	3 weeks x 2	LISS vs. HIIT	VO_{2max}	Lactate threshold, Hrsubmax, Wmax	all parameters = >2 S.E
					HIIT	Eight blocks of 170% of VO2max (20 seconds activity, 10 seconds rest), 4 sessions/week					
Ross [44]	2015	RCT	121	Male and female. Sedentary. Average age 53.2.	LALI	180-300kcal at 50% of VO2max per session, 5 sessions/week	24 weeks	LALI vs. HALI vs. HAHI at 4, 8, 16 and 24 week intervals	VO_{2max}	N/A	VO2max = >1 T.E.
					HALI	360-600kcal at 50% of VO2max per session, 5 sessions/week					

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					HAHI	360-600kcal at 75% of VO2max per session, 5 sessions/week					
Gurd [45]	2016	RCT	63	Healthy males who had previously attended prior, similar studies.	HIIT	Synthesis of five HIIT studies, each with differing methods.	4-6 weeks	HIIT vs. control	VO_{2max}	Lactate threshold, time to completion	VO2max = >2 T.E.
Yan [46]	2017	Cohort	39	Male only. Moderately-trained. 18-45 years old. Caucasian only. BMI < 30.	HIIT	Variable intensity tailored to individual's pre-intervention measures, 3 HIIT sessions/week	4 weeks	N/A (no control/comparison)	VO_{2max}	Lactate threshold, power output, distance	VO2max Δ > 0%
Scharhag-Rosenberger [47]	2012	Cohort	18	Male and female. 32-50 years old. BMI 19-28.	END	Jogging or walking, 45 minutes duration at intensity of 60% HR or HR at lactate threshold, 3 sessions/week	50 weeks	N/A (no control/comparison)	VO_{2max}	Lactate threshold, resting heart rate, submaximal heart rate	Variable; VO2max = Δ > 5.6%
Higgins [48]	2015	Cohort	23	No population data provided in detail.	HIIT	Cycling. 3 sessions/week	6 weeks	N/A (no control/comparison)	VO_{2max}	glucose, S&DBP, lipid profile	VO2max = >2 S.E.
Astorino [29]	2018	Cohort	14	Male and female. Healthy. Previously active (150 minute/week). 20-49 years old.	HIIT	8 to 10 1 minute rounds of HIIT, 130% of power output based on volunteer ventilatory threshold, 3 session/week	3 weeks	control vs. HIIT	VO_{2max}	Time trial performance, ventilatory threshold	VO2max = >2 S.E.

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Kohrt [26]	1991	Cohort	320	Male and female. Healthy. Untrained. 60-71 years old.	END	Up to 50 minutes/day, 85% of Hrmax, graded increase in volume and intensity up to third month, daily exercise	36-52 weeks	control vs. END	VO_{2max}	RERmax, Vemax, Hrmax, Hrrest	None
Nybo [18]	2010	Cohort	36	Male only. Untrained. No training in prior two years. 20-43 years old.	HIIT	Running. 20 minute total exercise duration. Five sets of two minutes at above 95% of calculated Hrmax. 2 sessions/week	12 weeks	HIIT vs. END vs. RET vs. control	VO2max	Hrrest, Bprest, Hrmax	None
					END	Treadmill. 60 minute at 80% of Hrmax. 2.5 sessions/week					
					RET	Lower body compound and isolation exercises, 3-4 sets per exercise, 1 minute rest interval, average duration 60 minutes, 2 sessions/week					
Osei-Tutu & Campagna [49]	2004	RCT	40	Male and female. Healthy. Sedentary. 20-40 years old.	END	"Long Bout" modality considered: 30 minutes/day, 60-	8 weeks	"long bout" (END) vs. "short bout" vs. control	VO2max	N/A	None

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						79% Hrmax, 5 sessions/week					
Trapp [50]	2008	RCT	45	Female only. Healthy. Inactive. 18-30 years old. Mixed ethnicity.	HIIT	Ergometer. 8s sprinting, 12s recovery. Maximum of 60 bouts per session. Up to 20 minutes/weesion. 45 total exercise sessions in intervention period.	15 weeks	HIIT vs. END vs. control	VO2max	Body fat and muscle comparisons	None
					END	Ergometer. Exercise at 60% VO2max. Up to 40 minutes exercise per session.					
Metcalfe [51]	2012	RCT	29	Male and female. Healthy. Sedentary. Range for intervention and control averages 19-26 years old.	HIIT	Ergometer, maximal pedalling against 7.5% volunteer bodyweight, 10 minute duration, 3 sessions/week	6 weeks	HIIT vs. control	VO2max	N/A	None
Ziemann [52]	2011	RCT	21	Male only. Healthy. Inactive. College age.	HIIT	Six 90s maximal effort & 180s recovery rounds, set to 80% VO2max, 3 sessions/week	6 weeks	HIIT vs. control	VO2max	Power output (multiple parameters)	None
Burgomaster [53]	2008	Cohort	20	Male and female. Healthy.	END	40-60 minutes cycling, 65% of	6 weeks	HIIT vs. END	VO2max	Hrmax, RER	None

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				intense exercise for at least 1 year prior. Range for intervention and control averages 23-24 years old.		VO2max, 5 sessions/week					
					HIIT	30s of 4-6 bouts (increasing count), maximum effort (Wingate test), 4.5 minute recovery interval, 3 sesssion/week					
Lo [54]	2011	RCT	34	Male only. Healthy. Inactive. Average age 20.4 years old.	END	30 minute treadmill, 70-85% Hrmax, 3 session/week	24 weeks	END vs. RET	Hrmax	N/A	None
					RET	10 exercises (full body, compound and isolation), incremental increase in intensity (weight), 3 session/week					
McKay [55]	2009	RCT	12	Male only. Healthy. Not previously in a formal training programme. Average age 25 years old.	END	90-120 minutes of 65% pre-training VO2max. Permitted 60-90s intermittent recovery if required. 8 sessions total.	3 weeks	END vs. HIIT	VO2max	Wmax, lactate threshold (multiple parameters)	None
					HIIT	60s 120% Wrmax followed by 60s recovery for 8-12 bouts, 8 sessions total					

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Dunham [56]	2012	RCT	15	Male and female. Healthy. Physically active. Average age range 20.2-21.3 years old.	END	Ergometer, 45 minutes of 60-70% VO2max, 3 sessions/week	4 weeks	END vs. HIIT	VO2max	RERmax, Hrmax	None
					HIIT	Ergometer, five bouts of 1 minute at 90% VO2max followed by 2 minute recovery, 3 sessions/week					
Macpherson [57]	2010	C	20	Male and female. Recreationally active only. Ages per groups 24.3±3.3 and 22.8±3.1.	END	Treadmill, continuous running, 30-60 minutes (progressive increase), at 65% VO2max intensity, 3 sessions/week	6 weeks	END vs. HIIT	VO2max	Cardiac output, resting metabolic rate	None
					HIIT	Treadmill, 30s maximal effort sprinting, 4 minute recovery interval, increase in bout count from 4 to 6, 3 sessions/week					
Shepherd [58]	2013	RCT	16	Male only. Sedentary. Healthy. Average age per groups 21±1 and 22±2.	END	Ergometer, 40-60 minutes per session (increasing volume during intervention), 65% VO2max, 5 sessions/week.	6 weeks	END vs. HIIT	Intramuscular triglyceride, perilipin	VO2max, Wmax, insulin and glucose	None

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					HIIT	Ergometer, 30s maximal effort cycling, 4.5 minute recovery interval with 30W resistance at 50rpm pace, increase in bout count from 4 to 6, 3 sessions/week.					
Warburton [59]	2004	RCT	20	Male only. Active. Healthy. Average age per total group 29±4.	END	Ergometer, 30-48 minutes per session (increasing volume during intervention), 64.3±3.7% VO2max, 3 sessions/week.	12 weeks	END vs. HIIT vs. control	Blood volume constituents	VO2max, Hrmax, Wmax, S&DBP	None
					HIIT	Ergometer, 2 minutes high effort (90% VO2max) followed by 2 minutes low effort (40% VO2max) for a number of bouts that would match the END equivalent in terms of total work output, 3 sessions/week.					
Berger [60]	2006	C	23	Male and female. No intense exercise in	END	Ergometer, 30 minute per session, intensity	6 weeks	END vs. HIIT vs. control	VO2max, time delay, primary amplitude,	Wmax, lactate threshold,	None

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				preceding two years. Healthy. Average age per total group 24±5.		60% VO2max, 3-4 sessions/week			primary time constant	Hrmax	
					HIIT	Ergometer, 20 1 minute bouts of 90% VO2max followed by 1 minute rest, 3-4 sessions/week					
Matsuo [61]	2013	RCT	42	Male only. Sedentary. Healthy. Average age in study 26.5±6.2.	END	Ergometer, 40 minutes, 60-65% VO2max, 60rpm maintained, 5 sessions/week	8 weeks	END vs. HIIT ("sprint") vs. HIIT ("HIAT")	VO2max, cardiac parameters (multiple)	Blood constituents	None
					HIIT	"sprint"; ergometer, 30s 120% VO2max >85rpm followed by 15s rest period for 7 rounds; "HIAT"; alternating between 3 minutes 80+% VO2max at 70-80rpm against 2 minutes 50% VO2max at 60rpm (cooldown at end for both); 5 sessions/week					
O'Donovan [62]	2005	RCT	42	Male only. Sedentary. Healthy. Age range 30 – 45	END (lo intensity)	Ergometer, 400kcal at 60% VO2max, 3 sessions/week	24 weeks	END (lower intensity) vs. END (higher intensity) vs. control	VO2max	Lipid profile, fibrinogen	None

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				years old.	END (hi intensity)	Ergometer, 400kcal at 80% VO2max, 3 sessions/week					
Sandvei [63]	2012	RCT	23	Male and female. Sedentary to moderately trained. Healthy. Age range 18-35 years old.	END	Outdoor running, 30-60 weeks (5 minute incremental increase per week), 70-80% Hrmax, 3 sessions/week	8 weeks	HIIT (sprint) vs. END	Glucose, insulin	Lipid profile, Hrmax, VO2max	None
					HIIT	Sprinting, 30s maximal effort followed by 3 minutes recovery, 5-10 sprints (1 incremental increase per week), 3 sessions/week					

Table 1: Study characteristic details of studies included

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Indicator	Modality	Datapoint count	IV	95% CI Lower	Upper
VO2max	END	8	5.59	3.84	7.35
	HIIT	7	3.54	0.78	6.31
	RET	1	-1	-3.69	1.69
Hrrest	END	2	-5.32	-10.9	0.25
	RET	1	-5	-7.25	-2.75
	HIIT	3	-4.6	-10.63	1.43
Hrmax	RET	1	4.1	-4.37	12.57
	END	5	-0.16	-3.01	2.7
	HIIT	4	-1.3	-6.03	3.43
PO	HIIT	4	39.23	-3.07	81.54
	END	2	35.49	-4.23	75.2
LT	END	1	0.6	-0.28	1.48
	HIIT	1	0.5	-0.21	1.21

Table 2 Rank order comparison between different training modalities

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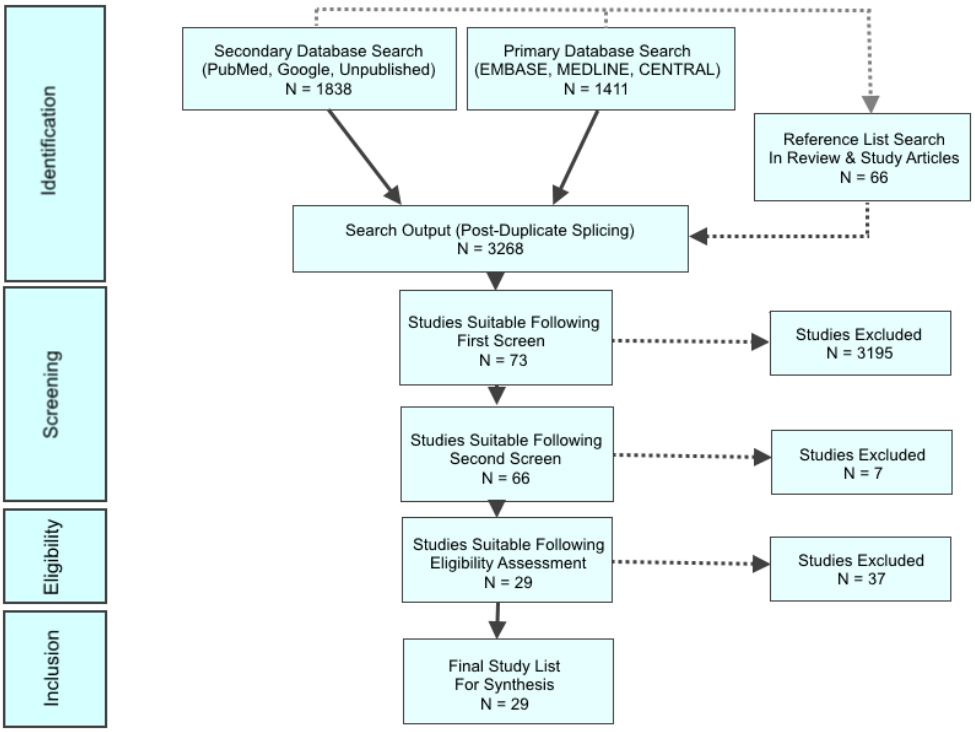
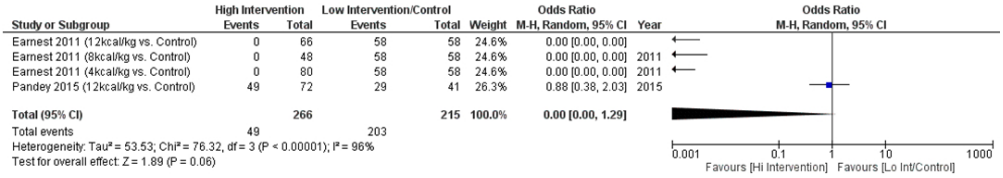


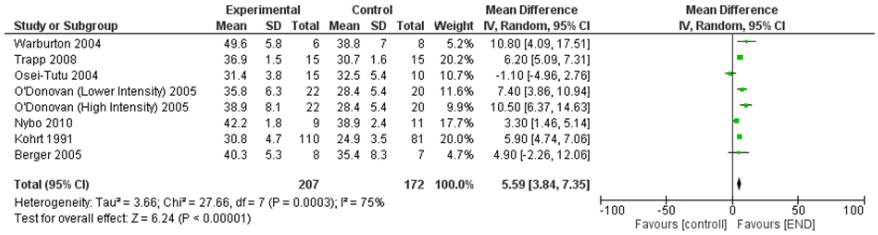
figure 1

215x162mm (96 x 96 DPI)

A



B



C

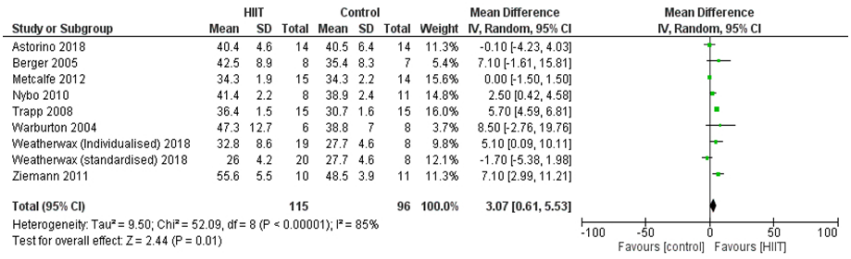
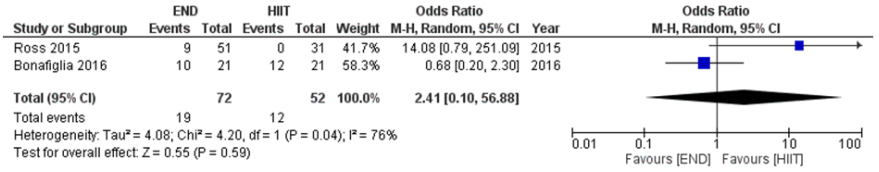


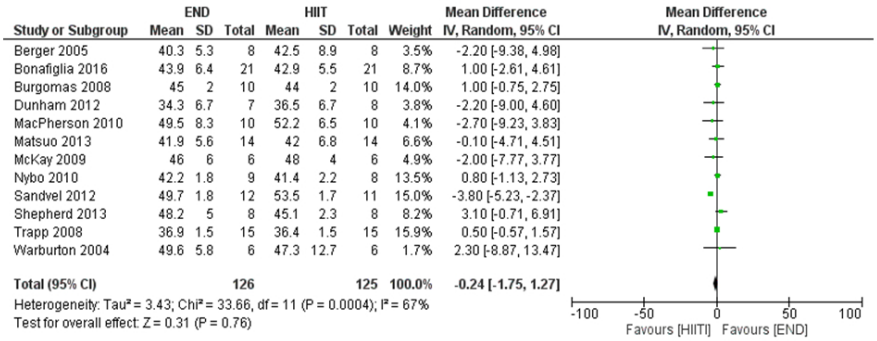
figure 2

285x291mm (96 x 96 DPI)

A



B



C

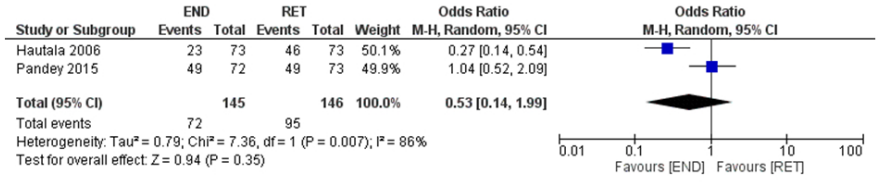


figure 3

285x291mm (96 x 96 DPI)

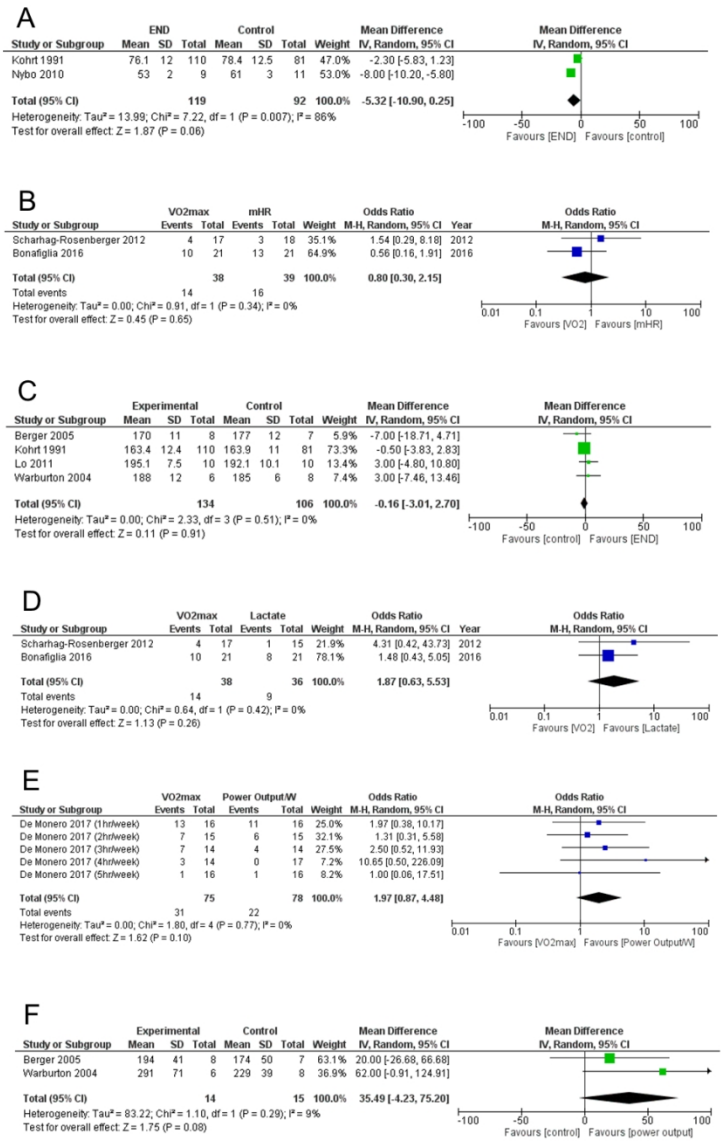


figure 4

285x423mm (96 x 96 DPI)

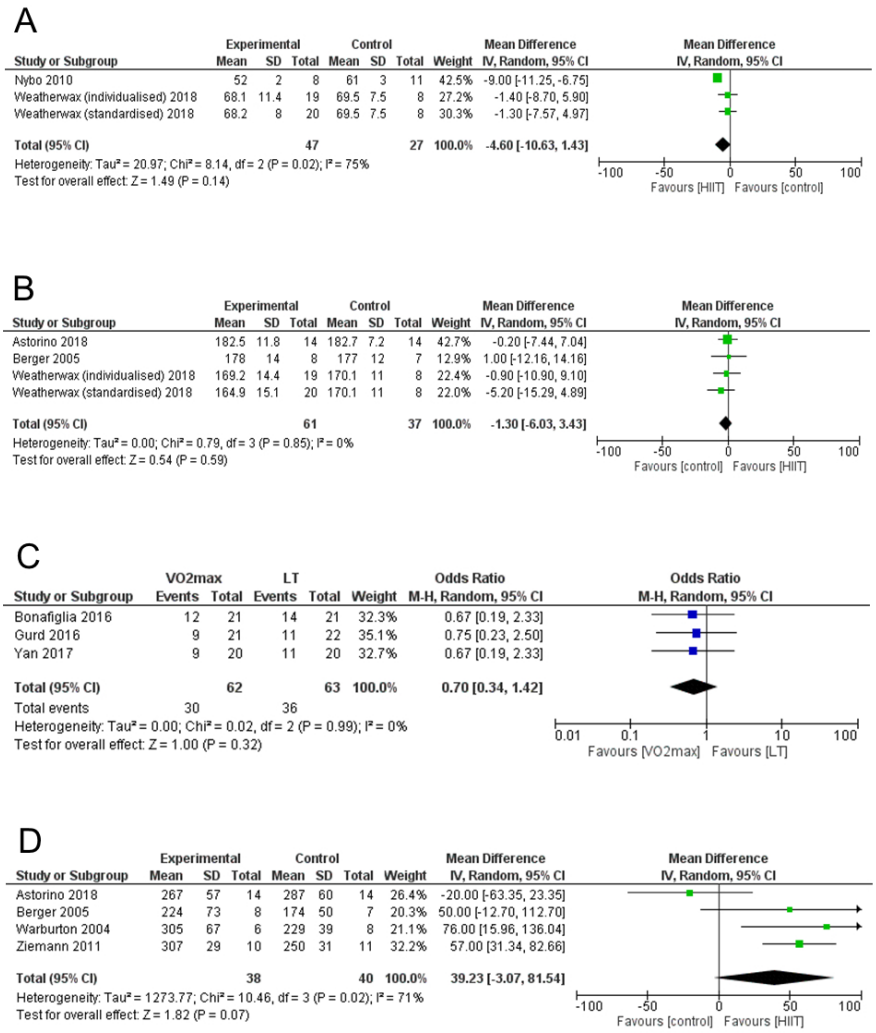


figure 5

285x343mm (96 x 96 DPI)

Medline (((“exercise” OR “training”) AND (“non-respon*” OR “non respon*” OR “no respon*” OR “nonrespon*”)) AND (“VO2*” OR “VO(2max)” OR “oxygen” OR “lactate” OR “cardiorespiratory fitness” OR “submaximal heart rate”)).ti,ab [Human age groups Young adult OR Adult OR Middle Aged] [Languages English] [Humans]

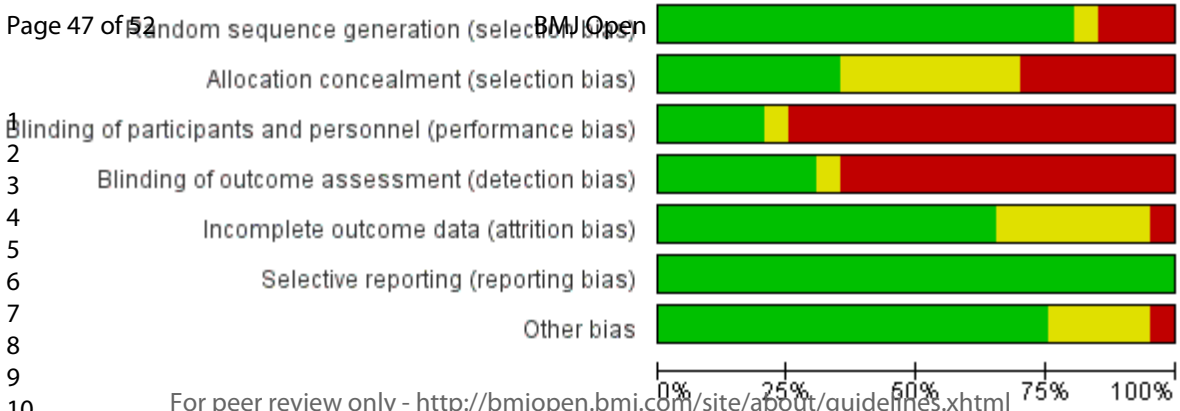
Viewing (66)

EMBASE (((“exercise” OR “training”) AND (“non-respon*” OR “non respon*” OR “no respon*” OR “nonrespon*”)) AND (“VO2*” OR “VO(2max)” OR “oxygen” OR “lactate” OR “cardiorespiratory fitness” OR “submaximal heart rate”)).ti,ab [English language] [Languages English] [Human age groups Adult 18 to 64 years OR Aged 65+ years] [Humans]

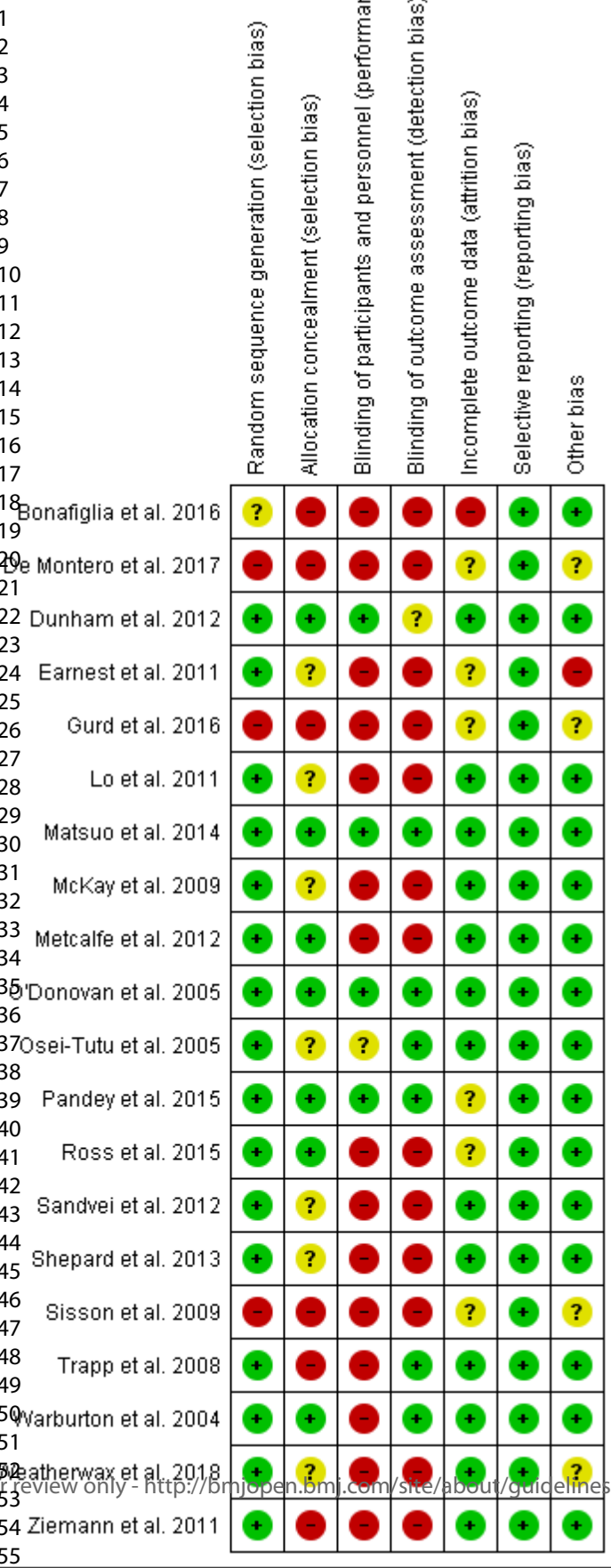
Viewing (110)

Figure 1 | MesH search strings used for the EMBASE and MEDLINE search strategies (performed on 22nd June 2019)

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Assessment of Risk of Bias (RoB) in case control and cohort studies (CLARITY framework)

Domain	Nybo et al. 2010	Yan et al. 2017	Scharhag-Rosenberger et al. 2012	Higgins et al. 2015	Astorino et al. 2018	Kohrt et al. 1991	Burgomaster et al. 2008	Macpherson et al. 2010	Berger et al. 2006
1. Can we be confident in the assessment of exposure?	Probable Low Risk	Low Risk	Probable Low Risk	Probable Low Risk	Low Risk	Probable High Risk	Low Risk	Probable Low Risk	Probable Low Risk
2.Can we be confident that cases had developed the outcomes of interest and controls had not?	Low Risk	Low Risk	Probable High Risk	Probable Low Risk	Probable Low Risk	Probable Low Risk	Probable Low Risk	Probable Low Risk	Probable Low Risk
3.Were the cases (those who were exposed and developed the outcome of interest) properly selected?	Probable High Risk	Probable Low Risk	Probable Low Risk	Probable Low Risk	Low Risk	Probable High Risk	Low Risk	Probable High Risk	Probable Low Risk
4.Were the controls (those who were exposed and did not develop the outcome of interest) properly selected?	Probable Low Risk	Probable Low Risk	Probable Low Risk	Probable High Risk	Probable Low Risk	Probable High Risk	Probable Low Risk	Probable High Risk	Probable Low Risk
5.Were cases and controls matched according to important prognostic variables or was statistical adjustment carried out for those variables?	Low Risk	Probable Low Risk	Probable Low Risk	Probable Low Risk	Probable Low Risk	Probable High Risk	Probable Low Risk	Probable Low Risk	Probable Low Risk

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Lead Author	Publication Date	Bias Domain				
		Risk of Bias	Imprecision	Inconsistency	Indirectness	Publication Bias
			RCTs no CI			
Sisson	2009	⊕⊕⊕⊕	reported	⊕⊕⊕	⊕⊕⊕⊕	not a systematic review
Earnest	2011	⊕⊕⊕⊕	⊕⊕⊕	⊕⊕⊕	⊕⊕⊕	not a systematic review
			no CI			
Pandey	2015	⊕⊕	reported	⊕⊕⊕	⊕⊕⊕	not a systematic review
Montero	2017	⊕⊕	⊕	⊕⊕⊕⊕	⊕⊕⊕	not a systematic review
			no CI			
Bonafiglia	2016	⊕⊕	reported	⊕⊕⊕⊕	⊕⊕⊕	not a systematic review
			no CI			
Ross	2015	⊕⊕⊕⊕	reported	⊕⊕⊕	⊕⊕⊕⊕	not a systematic review
			no CI			
Gurd	2016	⊕⊕⊕⊕	reported	⊕⊕⊕	⊕⊕⊕	⊕⊕
			no CI			
Yan	2017	⊕⊕⊕	reported	⊕⊕⊕	⊕⊕⊕	not a systematic review
Osei-Tutu &			no CI			
Campagna	2004	⊕⊕⊕	reported	⊕⊕⊕	⊕⊕	not a systematic review
			no CI			
Trapp	2008	⊕⊕⊕	reported	⊕⊕⊕⊕	⊕⊕	not a systematic review
			no CI			
Metcalfe	2012	⊕⊕⊕	reported	⊕⊕⊕	⊕⊕	not a systematic review
			no CI			
Ziemann	2011	⊕⊕⊕	reported	⊕⊕⊕	⊕	not a systematic review
			no CI			
Lo	2011	⊕⊕⊕	reported	⊕⊕⊕	⊕⊕	not a systematic review
			no CI			
McKay	2009	⊕⊕⊕	reported	⊕⊕⊕⊕	⊕⊕	not a systematic review
			no CI			
Dunham	2012	⊕⊕⊕	reported	⊕⊕	⊕⊕⊕	not a systematic review

Legend	
High	⊕⊕⊕⊕
Moderate	⊕⊕⊕
Low	⊕⊕
Very Low	⊕

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Shepherd	2013	⊕⊕	no CI reported	⊕⊕	⊕⊕	not a systematic review
Warburton	2004	⊕⊕	no CI reported	⊕⊕⊕	⊕⊕	not a systematic review
Matsuo	2013	⊕⊕	no CI reported	⊕⊕	⊕⊕	not a systematic review
O'Donovan	2005	⊕⊕⊕	no CI reported	⊕⊕⊕	⊕⊕	not a systematic review
Sandvei	2012	⊕⊕⊕	no CI reported	⊕⊕	⊕⊕	not a systematic review
Weatherwax	2018	⊕⊕	no CI reported	⊕⊕	⊕⊕⊕	not a systematic review
Cohort or Case Control						
Scharhag-Rosenberger	2012	⊕⊕⊕	no CI reported	⊕⊕⊕⊕	⊕⊕⊕	not a systematic review
Higgins	2015	⊕⊕⊕	no CI reported	⊕⊕⊕	⊕⊕⊕	not a systematic review
Astorino	2018	⊕⊕⊕	no CI reported	⊕⊕⊕⊕	⊕⊕⊕	not a systematic review
Hautala	2006	⊕⊕⊕	no CI reported	⊕⊕	⊕⊕	not a systematic review
Nybo	2010	⊕⊕	no CI reported	⊕⊕⊕	⊕⊕⊕	not a systematic review
Burgomaster	2008	⊕⊕⊕	no CI reported	⊕⊕⊕	⊕⊕	not a systematic review
Macpherson	2010	⊕⊕⊕	no CI reported	⊕⊕⊕	⊕⊕⊕	not a systematic review
Berger	2006	⊕⊕⊕	no CI reported	⊕⊕⊕⊕	⊕⊕⊕	not a systematic review

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	6-8
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	8
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	9-12
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	9
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	9
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	9-10
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	9-10
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	10-11
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	10-11
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	11-12
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	11
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	11

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	11-12
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	11
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	13
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	13
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	13
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	13-17
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	13-17
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	13
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	13-17
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	18
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	19-20
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	20
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	20

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Indicators of Response to Exercise Training: A Systematic Review & Meta-Analysis

Ardavani A[1], Aziz H[1], Phillips BE[1,2,] Doleman B[3], Ramzan I[1], Mozaffar B[1], Atherton PJ[1,2], Idris I[1,2,4]

1. MRC-Versus Arthritis Centre for Musculoskeletal Ageing Research, University of Nottingham, Royal Derby Hospital Centre, Derby, UK.
2. NIHR, Nottingham BRC, University of Nottingham.
3. Department of Anaesthesia, University Hospitals Derby and Burton NHS Foundation Trust, Derby, UK.
4. Department of Endocrinology and Diabetes, University Hospitals Derby and Burton NHS Foundation Trust, Derby, UK.

Address for correspondence:
Dr Iskandar Idris
University of Nottingham Medical School,
Royal Derby Hospital,
Uttoxeter Road, Derby,
DE22 3DT, UK.
Email: iskandar.idris@nottingham.ac.uk

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Abbreviations

BMI	Body mass index
CI	Confidence interval
CRF	Cardiorespiratory fitness
CSA	Cross sectional area
END	Endurance exercise training
GRADE	Grading of Recommendations, Assessment, Development and Evaluations
HIT	High intensity interval training
HR	Heart rate
HR_{max}	Maximum heart rate
HR_{rest}	Resting heart rate
IV	Inverse variance
LT	Lactate threshold
OR	Odds ratio
PO	Power output
QoL	Quality of life
RCT	Randomised control trial
RET	Resistance exercise training
RoB	Risk of bias
SD	Standard deviation
SE	Standard error
SMD	Standardised mean difference
SWC	Smallest worthwhile change
T2DM	Type 2 diabetes mellitus
VO_{2max}	Maximal rate of oxygen consumption
WHO	World Health Organisation

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Contributorship

Ardavani undertook data collection, analysis and wrote the first draft of the manuscript; Aziz undertook data collection and analysis; Phillips critically analysed data analysis, interpretation and provide technical advice on content; Doleman provided expertise in advanced meta-analysis technique; Mozaffar & Ramzan undertook bias assessment; Atherton: conceived the study and provided technical advice for content and writing; Idris: planned the study, provided overall supervision of the study conduct, analysis, interpretation and drafts. All authors contributed to the write up and approve the final draft.

Data sharing:

Additional data can be assessed by contacting the corresponding author.

Ethics:

Not applicable

Competing interests:

Ardavani A – None declared

Aziz H – None declared

Phillips BE – None declared

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Idris I – None declared

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ABSTRACT

Background: Means-based analysis of maximal rate of oxygen consumption (VO_{2max}) has traditionally been used as the exercise response indicator to assess the efficacy of endurance (END), high intensity interval (HIIT) and resistance exercise training (RET) for improving cardiorespiratory fitness (CRF) and whole-body health. However, considerable heterogeneity exists in the inter-individual variability response to the same or different training modalities.

Objectives: We performed a systematic review and meta-analysis to investigate exercise response rates in the context of VO_{2max}:

i) in each training modality (END, HIIT and RET) versus controls, ii) in END versus either HIIT or RET, and iii) exercise response rates as measured by VO_{2max} versus other indicators of positive exercise response in each exercise modality.

Methods: Three databases (EMBASE, MEDLINE, CENTRAL) and additional sources were searched. Both individual response rate and population average data was incorporated through continuous data, respectively. Of 3,268 identified manuscripts, a total of 29 studies were suitable for qualitative synthesis and a further 22 for quantitative. Stratification based on intervention duration (less than 12 weeks; more than or equal to 12 weeks) was undertaken.

Results: A total of 62 data-points were procured. Both END and HIIT training exhibited differential improvements in VO_{2max} based on intervention duration. VO_{2max} did not adequately differentiate between END and HIIT, irrespective of intervention length. Although none of the other exercise response indicators achieved statistical significance, LT and HR_{rest} demonstrated common trajectories in pooled and separate analyses between modalities. RET data was highly limited. Heterogeneity was ubiquitous across all analyses.

Conclusions: The potential for LT and HR_{rest} as indicators of exercise response requires further elucidation, in addition to the exploration of interventional and intrinsic sources of heterogeneity.

Keywords: Exercise response, Resistance Exercise, High Intensity Interval training, Endurance, VO2 max, Power output, meta-analysis

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Article summary:

Strength and Limitations:

Strength:

- Robust analysis and synthesis of available evidence (randomised-control trial (RCT), case-control and cohort).
- For continuous data, we utilised the generic inverse variance (IV) statistical method with the Random Effects model and established the outcome measure as the standardised mean difference.
- Stratification of results dependent on intervention duration was performed.

Limitations:

- Significant heterogeneity of studies (exercise response indicators, training modalities, assessment protocols, population being studied and analytical methods) limits the ability to undertake more extensive meta-analysis of available data, resulting in the majority of outputs demonstrating statistical non-significance.
- Analysis was substantially restricted due to currently-insufficient data for several alternative markers, requiring further characterisation through qualitative assessment.

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INTRODUCTION

Physical activity in humans has been recognised to confer a beneficial effect on health since the time of Hippocrates and Galen.[1,2] The World Health Organisation (WHO) defines physical activity as "any bodily movement produced by skeletal muscles that requires energy expenditure".[3] Physical activity has been shown to not only be cardio-protective,[4] but prospective data has demonstrated an inverse correlation between increased physical activity and all-cause mortality.[3,5,6] Over the years, regular physical activity has been implicated in the prevention or management of a considerable number of chronic diseases, including cancer.[7,8] Physical activity has also been actively implemented as an intervention for age-associated frailty, resulting in a marked improvement in the quality of life (QoL) of older individuals,[9-11] as well as improvements in the constellation of age-associated metabolic abnormalities which include dyslipidaemia, hyperglycaemia, hypertension and obesity.[12-16]

Within the literature concerning structured exercise training as a form of physical activity, and away from specialist athletic training regimes, three broad variants of exercise training modalities are commonly described: i) Endurance exercise training (END), ii) high intensity interval training (HIIT), and iii) resistance exercise training (RET).[17,18] Each of these training modalities is associated with a multitude of differing components, including content, volume, intensity, duration (training and recovery periods) and frequency.[19-22] As such, each modality is associated with distinct improvements in musculoskeletal, metabolic and/or cardio/vascular parameters.[6-8] Guidelines published by the UK Department of Health and Social Care in September 2019 advise that healthy adults should perform at least two instances of RET and a total of 150 minutes of moderate END on a weekly basis.[23] There is also some recognition of HIIT, as a reduced volume vigorous exercise can replace moderate END as long as this is accumulated in bouts of 10 minutes or more.

In terms of determining the efficacy of an exercise training programme, despite the aforementioned differences which will likely occur with different training regimes, an assessment of cardiorespiratory fitness (CRF) is often undertaken. In humans this is most commonly undertaken through assessment of the maximal rate of oxygen consumption (VO_{2max}), defined as the peak utilization of oxygen by metabolically active tissue.[24,25] Subsequently, the VO_{2max} plateau has become the gold standard for determining the maximal

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CRF of an individual, in addition to serving as a comparative marker of response following an exercise intervention.[26-30]

Although it is well-established that improvements in multiple health parameters are generally observed in humans following a period of exercise training,[27,31] the observance of individuals who do not demonstrate an improvement for a particular indicator above measurement error has emerged in multiple studies (often defined as "non-responders").[31,32] Estimated to constitute up to ~20% of any given population for the primary expected physiological adaptation (e.g. hypertrophy for RET and VO_{2max} or insulin sensitivity for END),[32] numerous explanations have emerged to try and describe this phenomenon, ranging from innate factors to poor compliance.[32,33] However, simple explanations based on baseline characteristics or training compliance/intensity do not appear to be able to fully explain the marked heterogeneity in exercise adaptation.[27] Furthermore, a subset of individuals demonstrate a worsening of a given indicator below measurement error, with some describing these individuals as "adverse responders". [31,34] If using this nomenclature, one study reported that an estimated 7% of people are described as being adverse responders in at least two parameters,[33] with commonly assessed variables (in addition to $VO_{2max)}$ including heart rate (HR), lactate threshold (LT) and power output (PO).[34,35]

It must however be acknowledged that the concept of non-responders, and certainly that of adverse responders, is not universally accepted with ongoing debate in the scientific community concerning the epistemological validity of this concept.[27,31] Some of this debate is centered on the definitions of these concepts, with the work of Mitchell *et al.* 2014 claiming there are no non-responders to RET as all individuals demonstrated at least one positive adaptive response of the many that were measured.[36] This definition of a non-responder is however strikingly different to that used by Phillips and colleagues in their work looking at molecular networks of exercise adaptation. In this study, non-responders were classed as those who did not demonstrate significant hypertrophy during 20-weeks RET, but may have displayed other improvements such as strength or vascular conductance. As such, a number of these individuals would have been non-responders in one study but not in another. This apparent uncertainty has been further exacerbated by the observation of a dissociation between VO_{2max} and other exercise response indicators (including blood lactate and

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maximum heart rate (HR_{max})) with END.[37] Furthermore, evidence of a disparity in indicator-based responses has been demonstrated in analyses of outcomes following RET exercise[38] with hypertrophic gains not necessarily representing changes in muscle function, for example.

Of the three aforementioned exercise modalities and with respect to improvements in CRF, END and HIIT are each recognised as having a significant and positive effect overall. [18,39] In contrast, the benefits of RET are distinct from this and are traditionally considered to be improvements in strength and skeletal muscle hypertrophy[18]. Although early evidence determined that RET did not confer any improvement in VO_{2max},[40] a more recent comprehensive assessment through a narrative systematic review comprising 17 studies, concluded that improvements in VO_{2max} may be observed with RET in previously untrained individuals irrespective of age.[38] As such, sedentary individuals would conceivably benefit from a concurrent improvement in both CRF and the acknowledged skeletal muscle-based improvements associated with RET.[38]

As a result of the current paradigm within the literature, there exists an uncertainty concerning the suitability of different interventions to elicit an exercise response. In addition to the academic consideration of what may constitute as a "response" to any particular intervention, a determination of whether alternative markers of response to VO_{2max} exist in each of the three modalities has not yet been undertaken through a systematic approach.

Therefore, through a combined systematic review and meta-analysis strategy, this study appraises the available evidence of studies in order to answer the following research questions:

1. In untrained human adults, is each exercise modality (END, HIIT and RET) more effective than controls in eliciting an improvement in CRF based on VO_{2max}?
2. In untrained human adults, is END more effective than either HIIT and RET in eliciting an improvement in CRF based on VO_{2max}?
3. In untrained human adults and per each exercise modality, do other measures of exercise response (HR_{rest}, HR_{max}, LT and PO) elicit a similar rate of exercise response when compared with VO_{2max}?

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METHOD

The population assessed in this study was human adults between the ages of 18 to 80 years. As the extent of exercise response is noted to be similar between both males and females,[26] both sexes were included in all analyses. The investigated intervention training modalities were END, HIIT and RET. To permit the investigation of the three research questions, the comparisons and outcome measures used were i) intervention (END, HIIT or RET) versus control using VO_{2max} as the response indicator, ii) END versus HIIT or RET using VO_{2max} as the response indicator, and iii) VO_{2max} versus HR_{rest}, HR_{max}, LT or PO in each of END, HIIT and RET. The outcome measures included in this study were continuous data (i.e. numerical values defined by a defined scale and range, such as heart rate or watts) represented by mean±standard deviation (SD) values. A complete PRISMA checklist is provided (Supplementary 1).

The inclusion criteria are all published clinical trials with an intervention component (randomised-control trial (RCT), case-control and cohort), as the investigated phenomenon (indicators of exercise response) require an actioned stimulus. In previously untrained healthy or obese individuals or individuals with type two diabetes mellitus (T2DM) without physical impediment, and the utilisation of VO_{2max} as an endpoint. The exclusion criteria are limited to qualitative studies, non-human studies, studies where participants are younger than 18 or older than 80 years of age, physically impaired individuals, pregnant volunteers, or participants with any established cardiovascular, renal, musculoskeletal, neurological, malignant or pulmonary disease.

Three electronic online literature databases (EMBASE, MEDLINE, CENTRAL) were utilised as primary data sources. Search strategies for EMBASE and MEDLINE were undertaken (Supplementary 2). The search strategy for CENTRAL was “VO2 exercise response”. To screen for clinical trials not captured in the above search strategies, and to definitively address the anticipated deficiency in RET intervention studies, two further search procedures were performed through PubMed. The search terms in each were “VO2 exercise response” and “resistance exercise response variability”. Only clinical trials were selected. Furthermore, grey literature sources were sought in addition to the above (Google and ClinicalTrials.Gov). The search phrase “exercise response VO2” was submitted to the Google search engine (147

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results, seven relevant studies). An additional search was made through ClinicalTrials.Gov (completed studies, adult, all sex, "exercise response" search string, only studies with preliminary or final results selected, 1308 results, six relevant studies). A further search through the reference lists of studies selected for synthesis was implemented to further address any undetected primary sources.

The MeSH terms undertaken in this study are provided (Supplementary 2). Two of the primary database sources (EMBASE, MEDLINE) were searched by two independent researchers (AA, HA), with CENTRAL, PubMed, Google, ClinicalTrials.Gov and reference list review searches and further selection stages being undertaken by one researcher (AA).

In the first screening procedure, all studies were initially assessed on their implementation of END, HIIT and/or RET as a primary intervention and their utilisation of at least one response indicator, where VO_{2max} was mandatory. In accordance with the findings of a meta-analysis published by Bacon *et al.* 2013 that determined VO_{2max} trainability is increased with prolonged intervention periods,[41] studies with a duration of less than two weeks were excluded from the analysis. Following the completion and compilation of the studies obtained from the preliminary data source searches, duplicates were highlighted and the novel studies were spliced into a new list. Thereafter, any new studies obtained from the other search strategies were added to a continuously updated version of this list. The extracted summary data for all studies that were deemed suitable for synthesis included the study title, primary author, year of publication, association with any other studies or trials, study design type, target population characterisation, assessed interventions, duration of intervention, intervention detail (including means of exercise, frequency, volume parameters and/or intensity), primary and secondary endpoints, criteria for exercise response (if applicable), data type (categorical versus continuous) and suitability for any combination of the intended analyses specific for each research question (I to III).

The values incorporated in our synthesis were absolute units of post-intervention values in the pertinent data fields per exercise modality or assessed group (either interventional or control). The data specific to response rate included exercise modality (END, HIIT and/or RET), the parameters utilised (VO_{2max}, HR_{rest}, HR_{max}, LT and/or PO) and the data type (continuous

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data). With the incorporated data type, the post-intervention mean and SD for intervention and control (or other intervention) groups were entered for comparison. The information per study was separately recorded in three tables, each pertaining to one of the three research questions established (Supplementary files 3a-c). One study (Gurd *et al.* 2016), containing pooled unweighted sample complete or subset data from five studies,[42] was incorporated as a single datapoint.

The Cochrane Risk of Bias (RoB) 2 template was implemented for randomised studies.[43] Qualitative assessment of the studies was undertaken by three researchers. Both study and outcome level outputs were produced. The resulting outputs were generated through Review Manager (RevMan) v5.3.5. Additionally, case-control studies were assessed using the CLARITY McMaster University Risk of Bias assessment framework. These were independently undertaken by three researchers (AA, BM, IR).

One principal summary measure was produced in this meta-analysis. The generic inverse variance (IV) statistical method was selected with the Random Effects model and established the outcome measure as a standardised mean difference (SMD) with a standard error (SE) calculation for the assessed groups based on Cochrane recommendations.[44] Further, a simplified pooled SD for the generation of the SMD was utilised in all instances.[45] All data was reported as IV values and 95% CI for all individual studies, where individual datapoints that were not eligible for pooled analyses having CI values derived through SE values as advocated by Cochrane.[46] With pooled data reported with the addition of Z data and p-values.

Forest plots were generated for all datasets which contained at least three data-points. All statistics and forest plots were produced with RevMan v.5.3.5. Data per study was manually entered into each of the variable listed. No *post hoc* data merging between studies was undertaken. Measures of statistical heterogeneity were calculated using the I^2 statistic through RevMan v.5.3.5 and are reported within the produced forest plots. In order to address anticipated heterogeneity within our dataset, a stratified approach based on age, intervention duration and/or weekly modality frequency will be considered. Further, subgrouping within forest plots based on the above was undertaken if at least two data-points

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were present. Certain studies permitted the inclusion of multiple groups separately for comparative purposes (Table 1, Supplementary files 3a-c).

In order to assess for publication bias, the Grading of Recommendations, Assessment, Development and Evaluations (GRADE) approach[47] was adopted on a study-specific level and was assessed by two independent researchers (AA, IR). Due to the limited number of studies per area of analysis, statistical assessment of publication bias, meta-regression and Trial Sequential Analysis was infeasible.

Patient and public involvement

Patients and/or the general public were not involved in undertaking and devising this systematic review and meta-analysis. No external groups, stakeholders or members of the public were involved in any element of the study's inception, planning, implementation or analysis.

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RESULTS

A total of 3,268 studies were generated from the identification stage. A total of 29 studies - 20 RCTs and nine case-control or cohort studies - were deemed suitable for inclusion in our qualitative and quantitative appraisal (Figure 1). [18,26-30,42,48-70,71-79] The publication dates ranged from 1991 to 2018, where the majority of the assessed studies were not associated with any other trials. A total sample size of 1,937 individuals was assessed.

The study characteristic data for all studies are presented (Table 1). The assessed populations were found to be heterogeneous with respect to biological sex, degree of physical fitness, body mass index (BMI) and age. The majority of studies featured a comparison between END and HIIT and included either END or HIIT within their analysis, with only three studies assessing the response rates for RET. One study[48] was found to employ a parallel assessment of exercise intensity and volume. The implemented exercise protocols exhibited substantial heterogeneity, with only two studies[49,50] displaying a congruent basis for their assessment due to their common derivation from the DREW study. The intervention period varied between three to 52 weeks. In accordance with our inclusion criteria, the majority of studies utilised VO_{2max} as a primary endpoint, with most studies also incorporating data pertaining to body composition. 12 of the 29 studies defined exercise responsiveness through differing thresholds of VO_{2max} change, where three of these 12 defined any improvement from VO_{2max} baseline ($\Delta > 0\%$) as evidence of a positive adaptation in their cohorts.[49,50,51]

The RoB summaries are provided (Supplementary files 4 - 6), which demonstrate varied degrees of bias across the assessed domains. Furthermore, GRADE score appraisal of each study was undertaken (Supplementary 4). A total of 62 data-points were obtained from these studies for inclusion in the forest plots (Figures 1-5). Sufficient overlap and representation in characteristics permitted a stratification based on intervention duration, with 12 weeks selected as the criteria for group formation.

Exercise Responsiveness versus Controls Using VO_{2max} (Analysis 1)

END vs. CON

Nine data-points from eight studies were observed (Supplementary 3a). Within the <12 week subgroup, the response through END as an intervention did not result in an intervention-favouring (IV = 0.06, 95% CI = -1.01 – 1.13) or statistically significant (p=0.64) outcome (Figure

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2). However, the ≥12 week subgroup demonstrated an unequivocal and statistically significant improvement in VO_{2max} (IV = 2.0, 95% CI = 0.68 – 3.32, p<0.05).

HIIT vs. CON

Nine data-points from eight studies were incorporated in this analysis (Figure 2). Although the effect size trends observed in both the <12 and ≥12 week subgroups demonstrated congruence with the duration-based observation in the END assessment (Figure 2), the results did not reach statistical significance (p=0.18-0.66) (Figure 2).

RET vs. CON

Only two studies (Nybo *et al.* 2010, Hautala *et al.* 2006) contained data indicating VO_{2max} improvements using RET (Supplementary 3a).[18,52] Neither data-points demonstrated an improvement in VO_{2max} through RET interventions lasting two and 12 weeks (IV = -0.35, 95% CI = -1.64 – 0.94; IV = 0, 95% CI = -1.43 – 1.43, respectively) (Supplementary 3a).

END versus HIIT and RET Responsiveness Using VO_{2max} (Analysis 2)

END vs. HIIT

Twelve studies provided data permitting a post-intervention comparison between END and HIIT intervention groups (Figure 3). The overall effect size in both the <12 and ≥12 week subgroups were found to be residual (IV = -0.29, 95% CI = -1.38 – 0.81; IV = 0.35, 95% CI = -0.12 – 0.81, respectively) and did not achieve statistical significance (Figure 3).

END vs. RET

Two studies (Nybo *et al.* 2010, Hautala *et al.* 2006) permitted a comparison between END and RET (Supplementary 3b).[18,52] Both studies demonstrated an improvement in END when compared to RET (IV = 1.62, 95% CI = 0.37 – 2.87; IV = 0.14, 95% CI = -1 – 1.28, respectively) (Supplementary 3b).

Exercise Responsiveness Using Other Exercise Response Indicators (Analysis 3)

When assessed with HR_{rest}, END demonstrated a consistent reduction when assessed through two studies (Kohrt *et al.* 1990, Nybo *et al.* 2010), where the duration was (or exceeded) 12 weeks (IV = -0.18, 95% CI = -1.92 – 1.56; IV = -3.14, 95% CI = -4.26 - -2.02, respectively) (Supplementary 3c).[18,26] Similarly, a pooled analysis demonstrated that interventions with HIIT exceeding (or lasting) 12 weeks demonstrated a reduction in following intervention (IV = -0.65, 95% CI = -4.29 – 0.99), although this result was not statistically significant (p=0.66) (Figure 5a). A single data-point (Nybo *et al.* 2010) was present for RET, which revealed a similar outcome (IV = -1.96, 95% CI = -3.1 - -0.82) (Supplementary 3c).[18]

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Five studies containing post-interventional data with respect to HR_{max} found no significant change in the effect size following END irrespective of training duration (<12 weeks; IV = 0.02, 95% CI = -1.92 – 1.96; ≥12 weeks; IV = 0.04, 95% CI = -1.43 – 1.50) (Figure 4a). A similar assessment using HIIT-based data demonstrated aligned outcomes (<12 weeks; IV = 0, 95% CI = -3.18 – 3.18, ≥12 weeks; IV = -0.23, 95% CI = -3.67 – 3.22) (Figure 4b). However, neither of these outcomes are statistically significant (Figure 4b). With respect to RET, Lo *et al.* 2011 and Hautala *et al.* 2006 revealed similar effect sizes, although the findings were also statistically non-significant (IV = 0.42, 95% CI = -3.81 – 4.65; IV = 0, 95% CI = -2.16 – 2.16, respectively) (Supplementary 3c).[52,53]

Within our dataset, only one study (Berger *et al.* 2006) presented data for END and HIIT pertaining to LT (Supplementary 3c). The effect sizes in both indicated a comparable improvement in LT following six weeks of training with either modality (END; IV = 0.68, 95% CI = 0.23 – 1.13; HIIT; IV = 0.71, 95% CI = 0.36 – 1.06) (Supplementary 3c).

Two studies (Warburton *et al.* 2005, Berger *et al.* 2006) contained data for PO changes following six and 12 weeks of END training, respectively (Supplementary 3c).[54,55] Both demonstrated a marginal improvement in PO (IV = 1.08, 95% CI = -28.93 – 31.09; IV = 0.44, 95% CI = -22.71 – 23.59, respectively) (Supplementary 3c). In HIIT, interventions of less than six weeks demonstrated an overall effect size that suggested an improvement (IV = 1.26, 95% CI = -9.17 – 11.69), although this result was not statistically significant (Figure 5b). No comparable data pertaining to improvements in PO following RET was observed in our dataset (Supplementary 3c).

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Discussion

This systematic review and meta-analysis investigated the variability in reported responses to END, HIIT and RET. We found that various factors such as training modality, training duration and response indicators may affect the reported exercise training responses. HIIT demonstrated a significant effect size using VO_{2max} versus controls. Overall, END resulted in a significant improvement in CRF versus controls using $VO_{2max,}$ but this was only statistically significant within our dataset examining training periods of 12 weeks or longer (Figure 2a,b). Although HIIT demonstrated a similar pattern, the results were not found to be statistically significant (Figure 2b). This finding contradicts an earlier meta-analysis performed in 2015, which assessed the responsiveness of END and HIIT in healthy adults between the ages of 18 to 45 years.[39] This difference may be due to our inclusion of studies including older cohorts, potentially reflecting an age-determined effect differential between exercise modalities. Age has previously been identified as a source of attenuation in exercise response – In 2010, Earnest *et al.* conducted an RCT study assessing 251 postmenopausal women over a 6 month period, where the intervention was exercise on a cycle ergometer at 4, 8, or 12 kcal/kg/week vs control (no exercise).[50] They were stratified into three groups based by age - <55 years, 55–59 years, >60 years. The results indicated that the control group had a reduction in maximal aerobic capacity by 1.6% (95% CI, −4.8, 1.0). Moreover, there was a reduced training response attributed to age and a correlation between age group and intervention ($P < 0.0002$).[50]

Likely a consequence of the historically-attested reduced effect afforded by RET in improving CRF, a deficiency in data concerning RET and exercise response indicators of CRF was observed which limits our ability to form comparisons with END and HIIT data. As a consequence of this, the currently-available data cannot elucidate the potential for an improvement in $VO2_{max}$ in untrained individuals through RET.[38] However, additional studies suggest a similar responsiveness to exercise when RET is compared to END. Pandley *et al.* 2015 conducted a RCT with 202 diabetics for nine months, where the interventions were aerobic training, resistance training or a combination of both.[30] The control group was a non-exercise group. The participants involved in exercise training were classified according to their ΔVO_{2max}, where fitness responders had a ΔVO_{2max} ≥5% and non-responders had a ΔVO_{2max} <5%. There were a similar proportion of fitness responders in the aerobic training–only (31.3%) and resistance training–only (33.9%) groups. Fitness non-responders had a Δ

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VO_{2max} −0.07 (95% CI, −0.1 to 0.04) and fitness responders had a ΔVO_{2max} 0.24 (95% CI, 0.20–0.28), p-value <0.001.[30]

Our findings concerning LT are partially in accordance with a recent cohort study in eleven moderately-trained cyclists which determined that PO exhibited a more pronounced relationship with athletic performance than $VO_{2max.}$[71] Our inconclusive outcome concerning LT appears to contradict earlier work, where a moderate positive correlation ($r^2 = 0.39$, $p < 0.05$) was inferred from a prior cohort study assessing the effect of END in sedentary males.[72] Additional studies demonstrated similar outcomes - Yan *et al.* 2017 conducted a multi-centre study where they sought to recruit 200 individuals, to determine the response to one session of high-intensity interval training (HIIT) and four weeks of HIIT.[51] In retrospect, they stated 39 individuals had done HIIT and found there was an average improvement in VO_{2max} of 3.85% ($p<0.001$) and an increase in lactate threshold (LT) of 9.01 ± 6.66% ($p<0.001$). Further, Gurd *et al.* 2016 used data from five previously published studies that included 63 adults, to ascertain the response to sprint interval training protocols. Responders for VO_{2max} was 41% and for LT was 50%.[42]

No data was assessed concerning the relationship between PO and RET, although this will ostensibly demonstrate an improvement due to the recognised development of skeletal muscle through type II fibre cross-sectional area (CSA) increases, sarcoplasmic hypertrophy and neuromuscular efficiency.[73]

A major limitation to our investigation is the ubiquitous heterogeneity in study design, intervention(s) and population characteristics were a recurrent feature in the assessed literature. A paucity in congruent measures restricted our ability to perform the appropriate multi-study sub-analysis. Indeed, exercise intensity is an established variable in the determination of exercise response, as defined by group-wide changes to VO_{2max}. This was demonstrated by Ross *et al.* 2015, who conducted a RCT with 121 individuals that completed a minimum 90% of five weekly exercise sessions over a 24-week period. Although there was an increase in cardiorespiratory fitness in all three groups at 24 weeks ($p<0.001$), An increase in the intensity of exercise (when matched for volume) resulted in a decrease in the number of cardiorespiratory non-responders.[48]

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Similarly, the majority of comparisons incorporated data from studies that implemented differing exercise training protocols. This heterogeneity is reflected in the statistical data, where statistical heterogeneity is demonstrated (I^2 > 50%) in the majority of the incorporated forest plots. Furthermore, exercise response is a complex trait, with multiple innate and environmental factors implicated.[74] As such, the absence of further participant differentiation due to the paucity in data, in combination with an enumeration of the known variables in exercise responsiveness ("trainability"), represents a source of confounding.[75]

The apparent heterogeneity described above likely contributes to the inability for all of the potential exercise response alternative indicators to achieve statistical significance (Figures 2-5). However, some consistency in outcomes were nonetheless observed in our analyses. HR_{max} did not demonstrate any clear direction in effect size in either END or HIIT (Figures 4a,b). Further to this, HR_{rest} was reliably reduced in END, HIIT and RET (Figure 4). The potential for PO to serve as an alternative to VO_{2max} remains inconclusive based on the presented data (Figure 5b), although our findings are not aligned with Montero and Lundby 2017, who carried out an RCT with 78 individuals over a six week period, where they performed 60 minute sessions per week of endurance training on a cycle ergometer.[28] The results from this study indicated that the more often exercise was performed, an improvement in both ΔVO_2max (p-value <0.001) and in ΔWmax (p<0.001) was observed.[28] As such, differing volume strategies and patient demographics between this study and those which achieved synthesis in our study are anticipated sources of confounding between these disparate results.

Differing criteria for exercise non-responsiveness in studies utilising VO_{2max} was observed, resulting in added variability. This limitation was also present with other indicators. Lastly, cardiovascular-dependent indicators of exercise response typically require longer to elicit a change in comparison to $VO_{2max.}$[76] As such, we speculate that HR_{rest} may serve as a reliable alternative for all exercise modalities, particularly over longer timeframes of intervention.

The lack of consistency with respect to the definition of exercise non-response[31] using VO_{2max} requires addressing.[77] The utilisation of Δ > 0%, which was implemented in three studies, in our opinion is inappropriate given the acknowledged issue of measurement error.[62] Bonafiglia

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et al. 2018 derived an alternative approach, using response CI and the smallest worthwhile change (SWC).[78] Further elucidation on the relationship between VO_{2max} and PO across the different exercise modalities in the general population may indicate differing patterns of responsiveness. Similarly, the relationship between extrinsic factors and exercise indicators may yield differing effects on chronic adaptation in each exercise modality. Additionally, the relationship between the role of exercise indicators beyond VO_{2max} and the interference effect (the frequently-observed diminishment of RET-specific adaptations to muscle size and function in a concurrent training setting) [79] may reveal novel or anticipatory patterns, which may predict this outcome. As such, future work investigating the potential relationship(s) and degrees of collinearity between intrinsic patient characteristics, intervention characteristics and the potential alternative exercise indicators of interest, preferentially through multivariate or meta-regression analyses, is advocated.

In addition to demonstrating current areas of uncertainty within the literature, the feasibility of alternatives to VO_{2max} for exercise response are tentatively substantiated through this work. Although safe, cardiopulmonary testing in physiological studies serves as an additional logistical consideration which is mitigated through the consideration of less-intensive measures, such as $HR_{rest.}$

In conclusion, our findings highlight the potential role of alternative indicators of exercise response in differing exercise modalities. Additionally, the constraints presented by extensive differences in study design, intervention type and duration, measurement variation and population characteristics require addressing in the literature. Our results suggest, dependent on the addressing of confounders, for HR_{rest} and LT to be explored further as viable alternatives to VO_{2max}.

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Tables & Figures

Table 1 – Study characteristics of the studies that achieved synthesis.

Figure 1 – PRISMA flow diagram of the study selection and identification process.

Figure 2 – a) END versus controls using VO_{2max} (<12 or ≥12 week subgrouping), b) HIIT versus controls using VO_{2max} (<12 or ≥12 week subgrouping)

Figure 3 – END versus HIIT using VO_{2max} (<12 or ≥12 week subgrouping)

Figure 4 – a) END versus controls using HR_{max} (<12 or ≥12 week subgrouping), b) HIIT versus controls using HR_{max} (<12 or ≥12 week subgrouping)

Figure 5 – a) HIIT versus controls using HR_{rest} (≥12 week subgrouping), b) HIIT versus controls using PO (<12 week subgrouping)

Supplementary Files

Supplementary 1 – PRISMA Checklist

Supplementary 2 – EMBASE search strategy

Supplementary 3 – Study Results (a) VO_{2max} response in intervention versus control groups per exercise modality, b) VO_{2max} response comparison between END versus HIIT and END versus RET, c) Per-exercise modality comparison in response with alternative response indicators)

Supplementary 4 – GRADE score results for all studies.

Supplementary 5 - RoB (RCTs) Category Results

Supplementary 6 – RoB (RCTs) Individual Results

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Study	Year	Design	Sample size	Patient characteristics	Intervention	Exercise Protocol (Modality, Intensity, Volume, Frequency)	Intervention Period	Comparison Type	Primary Endpoint	Secondary Endpoint	Response criteria
Weatherwax [65]	2018	RCT	39	Male and female. Less than 30 minutes activity in 3 days per week. 30 to 75 years old.	HIIT	Stationary bike, elliptical machine or treadmill. Stepwise progression in intensity depending on exercise group.	12 weeks	Standardised vs. Individualised	VO_{2max}	Physical activity, sitting time, Hrrest, Hrmax, body measurement parameters, dietary intake	VO_{2max} Δ > ME (4.7%)
Sisson [49]	2009	RCT	310	Female only (post-menopausal). Sedentary. 45-75 years old. multiple ethnicities.	END	Recumbent cycle ergonometer or treadmill (alternating); 50% baseline VO_{2max}, 3-4 sessions/week	24 weeks	4 vs. 8 vs. 12kcal/kg/week	VO_{2max}	N/A	VO_{2max} Δ > 0%
Earnest [50]	2011	RCT	251	Female only (post-menopausal). Sedentary. 45-75 years old. Caucasian only.	END	Recumbent cycle ergonometer or treadmill (alternating); 50% baseline VO_{2max}, 3-4 sessions/week	24 weeks	control vs. 4 vs. 8 vs. 12kcal/kg/week	VO_{2max}	N/A	VO_{2max} Δ > 0%
Pandley [30]	2015	RCT	202	Male and female. T2DM. 30-75 years old.	END	Treadmill; 50-80% VO_{2max}, 3 session/week	36 weeks	control vs. LISS vs. RET vs. both	VO_{2max}	HbA1c, body measurement parameters, S&DBPrest SBPeak, Hrrest, RER, insulin use	VO_{2max} = Δ > 5%
					RET	Full body training, compound movements (upper, lower, abdominal); 10-12 repetition range, 3 sessions/week					
Montero [28]	2017	RCT	78	Male only. Sedentary. Healthy. 18-35 years old.	END, HIIT, mixed	Recumbent cycle ergonometer; average of 65% of Wmax (between four intensity profiles), range from 1-5 sessions/week	6 weeks	Exercise frequency (1 vs. 2 vs. 3 vs. 4 vs. 5 sessions/week)	VO_{2max}, W_{max}	haemoglobin mass, plasma volume, red blood cell volume, blood volume, body measurement parameters, mitochondrial volume density	VO_{2max} & Wmax = >1 T.E.
Bonafiglia [27]	2016	RCT (crossover)	21	Male and female. Recreationally active	END	30 minutes cycling at 65% of VO_{2max}, 4 sessions/week	3 weeks x 2	LISS vs. HIIT	VO_{2max}	Lactate threshold, Hrsubmax, Wmax	all parameters =

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				only. Age range not stated.	HIIT	Eight blocks of 170% of VO_{2max} (20 seconds activity, 10 seconds rest), 4 sessions/week					>2 S.E
Ross [48]	2015	RCT	121	Male and female. Sedentary. Average age 53.2.	LALI	180-300kcal at 50% of VO_{2max} per session, 5 sessions/week	24 weeks	LALI vs. HALI vs. HAHI at 4, 8, 16 and 24 week intervals	VO_{2max}	N/A	VO_{2max} = >1 T.E.
					HALI	360-600kcal at 50% of VO_{2max} per session, 5 sessions/week					
					HAHI	360-600kcal at 75% of VO_{2max} per session, 5 sessions/week					
Gurd [42]	2016	RCT	63	Healthy males who had previously attended prior, similar studies.	HIIT	Synthesis of five HIIT studies, each with differing methods.	4-6 weeks	HIIT vs. control	VO_{2max}	Lactate threshold, time to completion	VO_{2max} = >2 T.E.
Yan [51]	2017	Cohort	39	Male only. Moderately-trained. 18-45 years old. Caucasian only. BMI < 30.	HIIT	Variable intensity tailored to individual's pre-intervention measures, 3 HIIT sessions/week	4 weeks	N/A (no control/comparison)	VO_{2max}	Lactate threshold, power output, distance	VO_{2max} Δ > 0%
Scharhag-Rosenberger [66]	2012	Cohort	18	Male and female. 32-50 years old. BMI 19-28.	END	Jogging or walking, 45 minutes duration at intensity of 60% HR or HR at lactate threshold, 3 sessions/week	50 weeks	N/A (no control/comparison)	VO_{2max}	Lactate threshold, resting heart rate, submaximal heart rate	Variable; VO_{2max} = Δ > 5.6%
Higgins [67]	2015	Cohort	23	No population data provided in detail.	HIIT	Cycling. 3 sessions/week	6 weeks	N/A (no control/comparison)	VO_{2max}	glucose, S&DBP, lipid profile	VO_{2max} = >2 S.E.
Astorino [29]	2018	Cohort	14	Male and female. Healthy. Previously active (150	HIIT	8 to 10 1 minute rounds of HIIT, 130% of power output based on volunteer	3 weeks	control vs. HIIT	VO_{2max}	Time trial performance, ventilatory threshold	VO_{2max} = >2 S.E.

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				minute/week). 20-49 years old.		ventilatory threshold, 3 session/week					
Kohrt [26]	1991	Cohort	320	Male and female. Healthy. Untrained. 60-71 years old.	END	Up to 50 minutes/day, 85% of HR_{max} graded increase in volume and intensity up to third month, daily exercise	36-52 weeks	control vs. END	VO_{2max}	RER_{max}, VE_{max}, HR_{max}, HR_{rest}	None
Nybo [18]	2010	Cohort	36	Male only. Untrained. No training in prior two years. 20-43 years old.	HIIT	Running. 20 minute total exercise duration. Five sets of two minutes at above 95% of calculated HR_{max} 2 sessions/week	12 weeks	HIIT vs. END vs. RET vs. control	VO_{2max}	HR_{rest}, BP_{rest}, HR_{max}	None
					END	Treadmill. 60 minute at 80% of HR_{max} 2.5 sessions/week					
					RET	Lower body compound and isolation exercises, 3-4 sets per exercise, 1 minute rest interval, average duration 60 minutes, 2 sessions/week					
Osei-Tutu & Campagna [68]	2004	RCT	40	Male and female. Healthy. Sedentary. 20-40 years old.	END	"Long Bout" modality considered: 30 minutes/day, 60-79% HR_{max} 5 sessions/week	8 weeks	"long bout" (END) vs. "short bout" vs. control	VO_{2max}	N/A	None
Trapp [69]	2008	RCT	45	Female only. Healthy. Inactive. 18-30 years old. Mixed ethnicity.	HIIT	Ergometer. 8s sprinting, 12s recovery. Maximum of 60 bouts per session. Up to 20 minutes/session. 45 total exercise sessions in intervention period.	15 weeks	HIIT vs. END vs. control	VO_{2max}	Body fat and muscle comparisons	None

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					END	Ergometer. Exercise at 60% VO_{2max}. Up to 40 minutes exercise per session.					
Metcalfe [70]	2012	RCT	29	Male and female. Healthy. Sedentary. Range for intervention and control averages 19-26 years old.	HIIT	Ergometer, maximal pedalling against 7.5% volunteer bodyweight, 10 minute duration, 3 sessions/week	6 weeks	HIIT vs. control	VO_{2max}	N/A	None
Ziemann [71]	2011	RCT	21	Male only. Healthy. Inactive. College age.	HIIT	Six 90s maximal effort & 180s recovery rounds, set to 80% VO_{2max}, 3 sessions/week	6 weeks	HIIT vs. control	VO_{2max}	Power output (multiple parameters)	None
Burgomaster [72]	2008	Cohort	20	Male and female. Healthy. Sedentary. No intense exercise for at least 1 year prior. Range for intervention and control averages 23-24 years old.	END	40-60 minutes cycling, 65% of VO_{2max}, 5 sessions/week	6 weeks	HIIT vs. END	VO_{2max}	HR_{max}, RER	None
					HIIT	30s of 4-6 bouts (increasing count), maximum effort (Wingate test), 4.5 minute recovery interval, 3 session/week					
Lo [53]	2011	RCT	34	Male only. Healthy. Inactive. Average age 20.4 years old.	END	30 minute treadmill, 70-85% HR_{max}, 3 session/week	24 weeks	END vs. RET	HR_{max}	N/A	None
					RET	10 exercises (full body, compound and isolation), incremental increase in intensity (weight), 3 session/week					
McKay [73]	2009	RCT	12	Male only. Healthy. Not previously in a formal training programme. Average age 25 years old.	END	90-120 minutes of 65% pre-training VO_{2max}. Permitted 60-90s intermittent recovery if required. 8 sessions total.	3 weeks	END vs. HIIT	VO_{2max}	W_{max}, lactate threshold (multiple parameters)	None

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					HIIT	60s 120% W_{max} followed by 60s recovery for 8-12 bouts, 8 sessions total					
Dunham [74]	2012	RCT	15	Male and female. Healthy. Physically active. Average age range 20.2-21.3 years old.	END	Ergometer, 45 minutes of 60-70% VO_{2max}, 3 sessions/week	4 weeks	END vs. HIIT	VO_{2max}	RER_{max}, HR_{max}	None
					HIIT	Ergometer, five bouts of 1 minute at 90% VO_{2max} followed by 2 minute recovery, 3 sessions/week					
Macpherson [75]	2010	C	20	Male and female. Recreationally active only. Ages per groups 24.3±3.3 and 22.8±3.1.	END	Treadmill, continuous running, 30-60 minutes (progressive increase), at 65% VO_{2max} intensity, 3 sessions/week	6 weeks	END vs. HIIT	VO_{2max}	Cardiac output, resting metabolic rate	None
					HIIT	Treadmill, 30s maximal effort sprinting, 4 minute recovery interval, increase in bout count from 4 to 6, 3 sessions/week					
Shepherd [76]	2013	RCT	16	Male only. Sedentary. Healthy. Average age per groups 21±1 and 22±2.	END	Ergometer, 40-60 minutes per session (increasing volume during intervention), 65% VO_{2max}, 5 sessions/week.	6 weeks	END vs. HIIT	Intramuscular triglyceride, perilipin	VO_{2max}, W_{max}, insulin and glucose	None
					HIIT	Ergometer, 30s maximal effort cycling, 4.5 minute recovery interval with 30W resistance at 50rpm pace, increase in bout count from 4 to 6, 3 sessions/week.					

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Warburton [54]	2004	RCT	20	Male only. Active. Healthy. Average age per total group 29±4.	END	Ergometer, 30-48 minutes per session (increasing volume during intervention), 64.3±3.7% VO_{2max}, 3 sessions/week.	12 weeks	END vs. HIIT vs. control	Blood volume constituents	VO_{2max}, HR_{max}, W_{max}, S&DBP	None
					HIIT	Ergometer, 2 minutes high effort (90% VO_{2max}) followed by 2 minutes low effort (40% VO_{2max}) for a number of bouts that would match the END equivalent in terms of total work output, 3 sessions/week.					
Berger [55]	2006	C	23	Male and female. No intense exercise in preceding two years. Healthy. Average age per total group 24±5.	END	Ergometer, 30 minute per session, intensity 60% VO_{2max}, 3-4 sessions/week	6 weeks	END vs. HIIT vs. control	VO_{2max}, time delay, primary amplitude, primary time constant	W_{max}, lactate threshold, HR_{max}	None
					HIIT	Ergometer, 20 1 minute bouts of 90% VO_{2max} followed by 1 minute rest, 3-4 sessions/week					
Matsuo [77]	2013	RCT	42	Male only. Sedentary. Healthy. Average age in study 26.5±6.2.	END	Ergometer, 40 minutes, 60-65% VO_{2max}, 60rpm maintained, 5 sessions/week	8 weeks	END vs. HIIT ("sprint") vs. HIIT ("HIAT")	VO_{2max}, cardiac parameters (multiple)	Blood constituents	None
					HIIT	"sprint"; ergometer, 30s 120% VO_{2max} >85rpm followed by 15s rest period for 7 rounds; "HIAT"; alternating between 3 minutes 80+% VO_{2max} at 70-80rpm against 2 minutes 50% VO_{2max} at					

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						60rpm (cooldown at end for both); 5 sessions/week					
O'Donovan [78]	2005	RCT	42	Male only. Sedentary. Healthy. Age range 30 – 45 years old.	END (lo intensity)	Ergometer, 400kcal at 60% VO_{2max}, 3 sessions/week	24 weeks	END (lower intensity) vs. END (higher intensity) vs. control	VO_{2max}	Lipid profile, fibrinogen	None
					END (hi intensity)	Ergometer, 400kcal at 80% VO_{2max}, 3 sessions/week					
Sandvei [79]	2012	RCT	23	Male and female. Sedentary to moderately trained. Healthy. Age range 18-35 years old.	END	Outdoor running, 30-60 weeks (5 minute incremental increase per week), 70-80% Hrmax, 3 sessions/week	8 weeks	HIIT (sprint) vs. END	Glucose, insulin	Lipid profile, HR_{max}, VO_{2max}	None
					HIIT	Sprinting, 30s maximal effort followed by 3 minutes recovery, 5-10 sprints (1 incremental increase per week), 3 sessions/week					
Hautala [52]	2005	RCT (crossover)	91	Male and female. Sedentary. Healthy. Average age in study 42±5.	RET	15 exercises including major muscle groups, 1 set of 8-12 repetitions to near fatigue. Resistance training every two days with a focus on arm or leg strength. 5 consecutive sessions/week	2 weeks	END vs. RET vs. control	BMI	VO_{2max}, HR_{max}, RER_{max}, maximum quadricep strength	VO_{2max} Δ > 0%
					END	40 minute cycle ergometer, 5 consecutive sessions/week					

Table 1 | Study characteristics of the studies that achieved synthesis.

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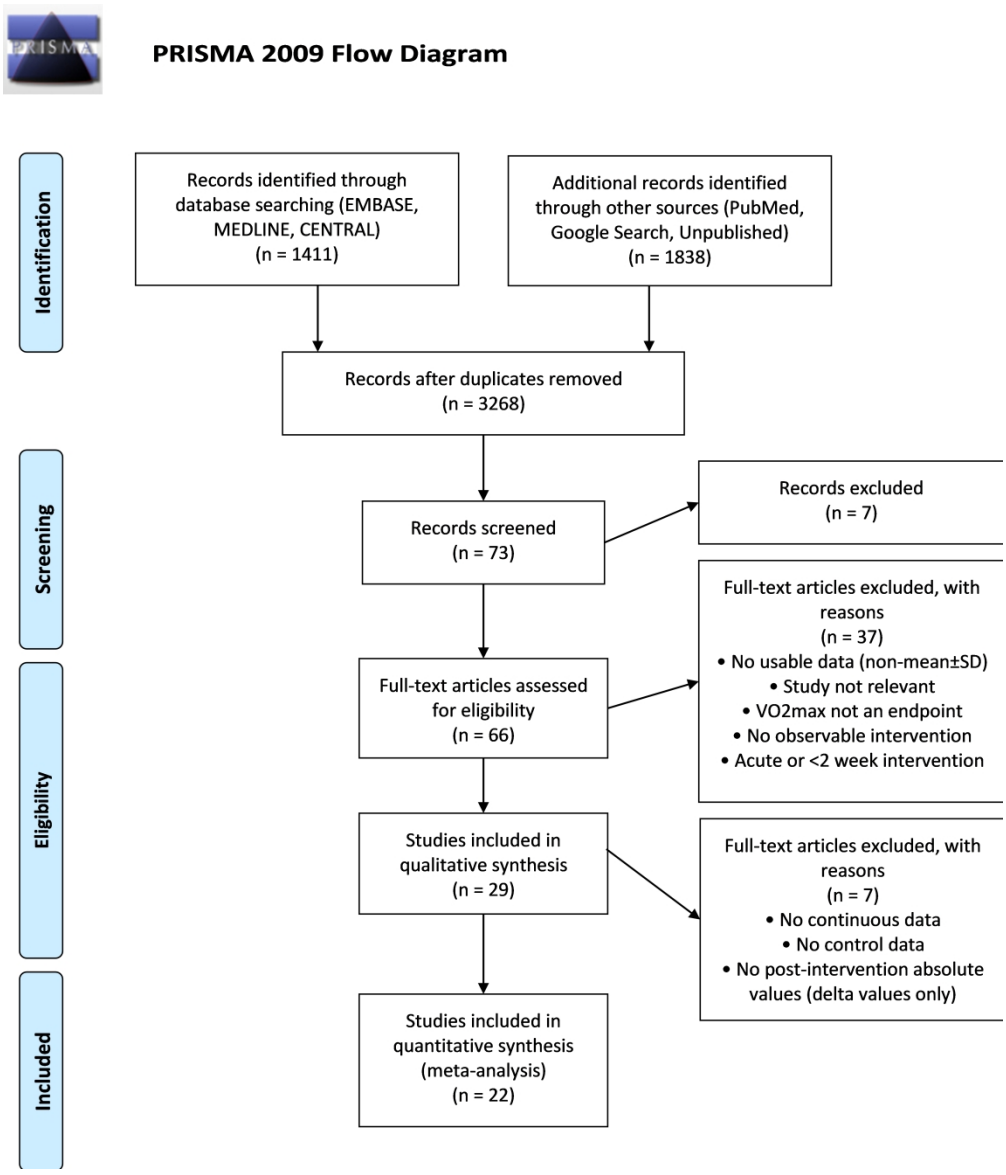
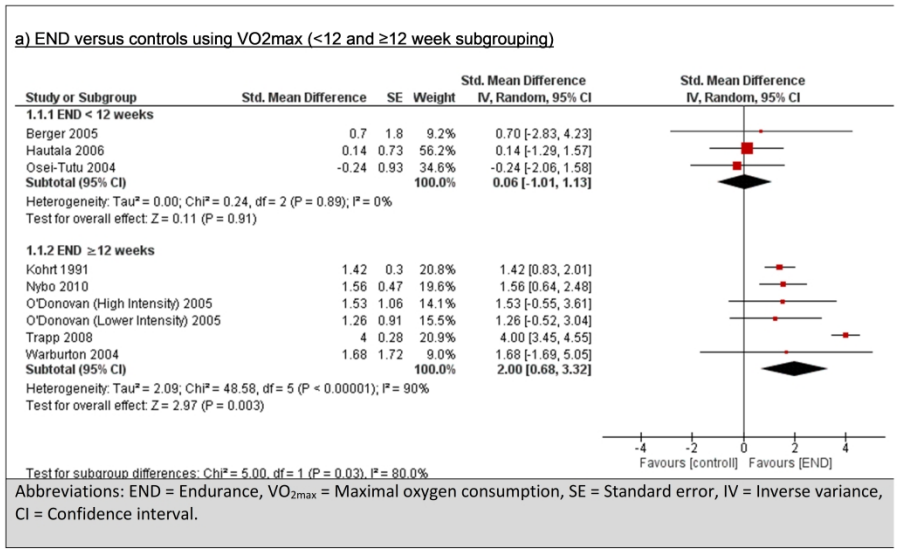
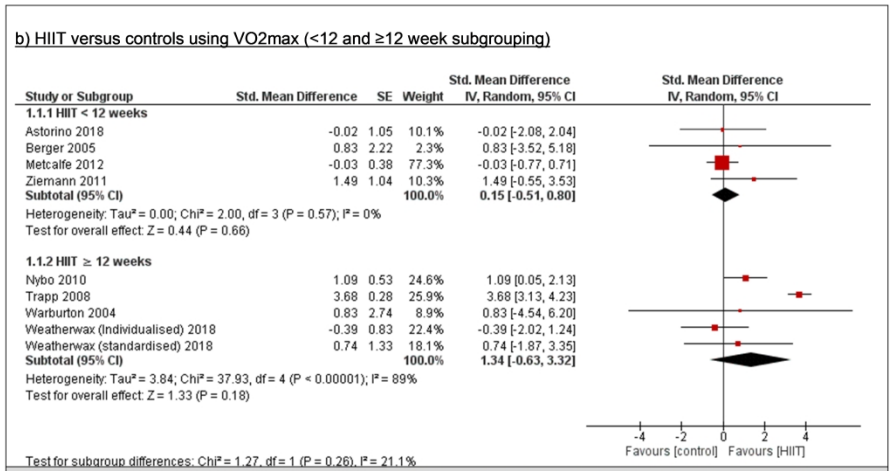


Figure 1

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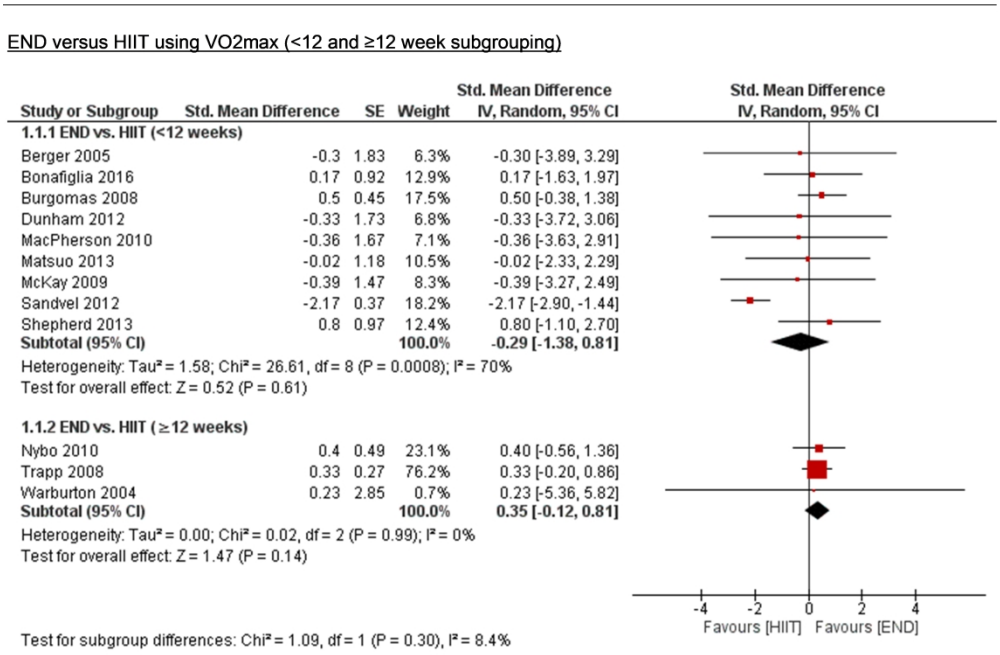
Abbreviations: END = Endurance, VO_{2max} = Maximal oxygen consumption, SE = Standard error, IV = Inverse variance, CI = Confidence interval.



Abbreviations: HIIT = High-intensity interval training, VO_{2max} = Maximal oxygen consumption, SE = Standard error, IV = Inverse variance, CI = Confidence interval.

Figure 2

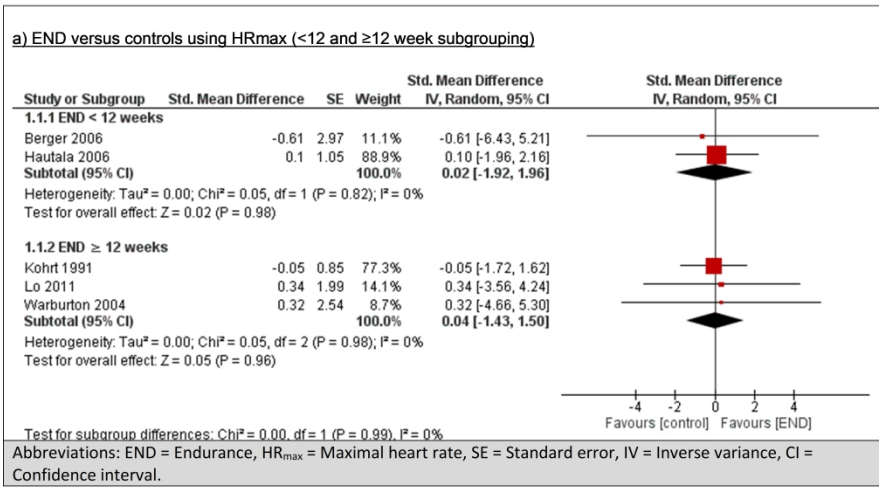
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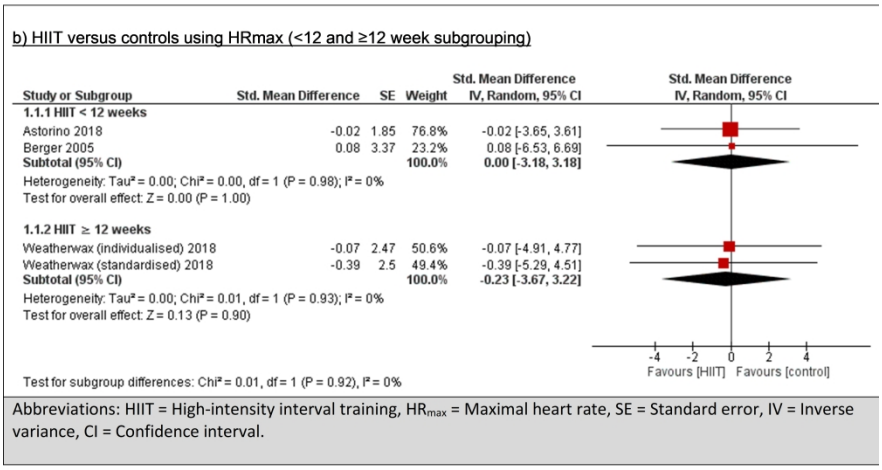
Abbreviations: END = Endurance, VO_{2max} = Maximal oxygen consumption, HIIT = High-intensity interval training, SE = Standard error, IV = Inverse variance, CI = Confidence interval.

Figure 3

1164x1164mm (96 x 96 DPI)



Abbreviations: END = Endurance, HR_{max} = Maximal heart rate, SE = Standard error, IV = Inverse variance, CI = Confidence interval.



Abbreviations: HIIT = High-intensity interval training, HR_{max} = Maximal heart rate, SE = Standard error, IV = Inverse variance, CI = Confidence interval.

Figure 4

1164x1593mm (96 x 96 DPI)

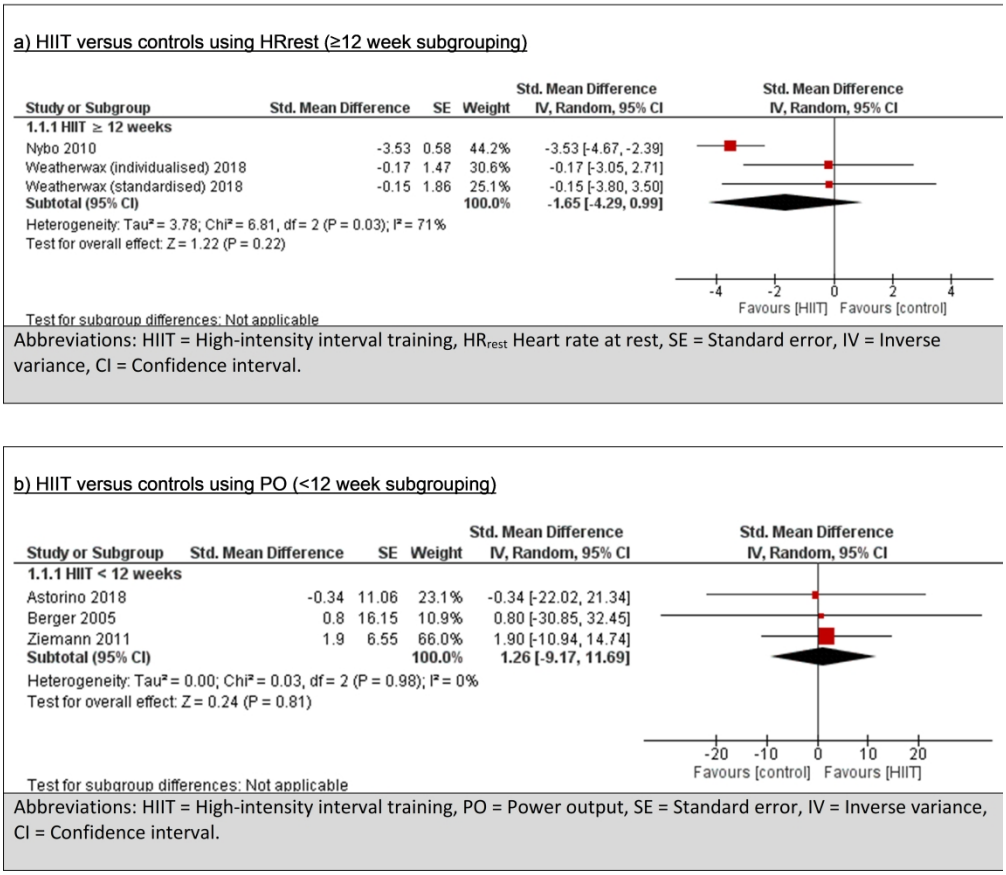


Figure 5

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Viewing (110)

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EMBASE (((“exercise” OR “training”) AND (“non-respon*” OR “non respon*” OR “no respon*” OR “nonrespon*”)) AND (“VO2*” OR “VO(2max)” OR “oxygen” OR “lactate” OR “cardiorespiratory fitness” OR “submaximal heart rate”)).ti,ab [English language] [Languages English] [Human age groups Adult 18 to 64 years OR Aged 65+ years] [Humans]

Viewing (110)

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Figure 3a | VO2max response in intervention versus control groups per exercise modality

Author	Date	Study Design Type	Intervention Duration	Intervention	Indicator	INT			CON			Derivative Statistics				95% CI	
						Mean	SD	Sample Size	Mean	SD	Sample Size	Pooled SD	SMD	SE	IV	Lower	Upper
Kohrt*	1991	C	36-52 weeks	END	VO2max	30.8	4.7	110	24.9	3.5	81	4.14	1.42	0.30	1.42	0.83	2.01
Nybo	2010	C	12 weeks	END	VO2max	42.2	1.8	9	38.9	2.4	11	2.12	1.56	0.47	1.56	0.64	2.48
Osei-Tutu & Campagna*	2004	RCT	8 weeks	END	VO2max	31.4	3.8	15	32.5	5.4	10	4.66	-0.24	0.93	-0.24	-2.06	1.58
Trapp	2008	RCT	15 weeks	END	VO2max	36.9	1.5	15	30.7	1.6	15	1.55	4.00	0.28	4.00	3.45	4.55
Warburton	2004	RCT	12 weeks	END	VO2max	49.6	5.8	6	38.8	7	8	6.43	1.68	1.72	1.68	-1.69	5.05
Berger	2005	C	6 weeks	END	VO2max	40.3	5.3	8	35.4	8.3	7	6.96	0.70	1.80	0.70	-2.83	4.23
O-Donovan (higher intensity)	2005	RCT	24 weeks	END	VO2max	38.9	8.08	22	28.4	5.4	20	6.87	1.53	1.06	1.53	-0.55	3.61
O-Donovan (lower intensity)	2005	RCT	24 weeks	END	VO2max	35.8	6.3	22	28.4	5.4	20	5.87	1.26	0.91	1.26	-0.52	3.04
Hautala	2006	RCT (crossover)	2 weeks	END	VO2max	37	7	73	36	7	18	7.00	0.14	0.73	0.14	-1.29	1.57
Nybo	2010	C	12 weeks	HIIT	VO2max	41.4	2.2	8	38.9	2.4	11	2.30	1.09	0.53	1.09	0.05	2.13
Trapp	2008	RCT	15 weeks	HIIT	VO2max	36.4	1.5	15	30.7	1.6	15	1.55	3.68	0.28	3.68	3.13	4.23
Metcalfe*	2012	RCT	6 weeks	HIIT	VO2max	34.3	1.9	15	34.3	2.2	14	2.05	-0.03	0.38	-0.03	-0.77	0.71
Ziemann	2011	RCT	6 weeks	HIIT	VO2max	55.6	5.5	10	48.5	3.9	11	4.77	1.49	1.04	1.49	-0.55	3.53
Warburton	2004	RCT	12 weeks	HIIT	VO2max	47.3	12.7	6	38.8	7	8	10.25	0.83	2.74	0.83	-4.54	6.20
Berger	2005	C	6 weeks	HIIT	VO2max	42.5	8.9	8	35.4	8.3	7	8.61	0.83	2.22	0.83	-3.52	5.18
Astorino **	2018	C	3 weeks	HIIT	VO2max	40.4	4.6	14	40.5	6.4	14	5.57	-0.02	1.05	-0.02	-2.08	2.04
Weatherwax (Standardised)	2018	RCT	12 weeks	HIIT	VO2max	26	4.2	20	27.7	4.6	8	4.40	-0.39	0.83	-0.39	-2.02	1.24
Weatherwax (Individualised)	2018	RCT	12 weeks	HIIT	VO2max	32.8	8.6	19	27.7	4.6	8	6.90	0.74	1.33	0.74	-1.87	3.35
Nybo	2010	C	12 weeks	RET	VO2max	37.9	3.3	8	38.9	2.4	11	2.89	-0.35	0.66	-0.35	-1.64	0.94
Hautala	2006	RCT (crossover)	2 weeks	RET	VO2max	36	7	73	36	7	18	7.00	0.00	0.73	0.00	-1.43	1.43

Legend

Notes

** Pooled mean and SD calculated between male and female groups. Reported to 1 d.p. ** Derived from RAMP (incremental exercise test) data.*

Abbreviations

C = Case-control, RCT = Randomised controlled trial, END = Endurance, HIIT = High-intensity interval training, RET = Resistance training, INT = Intervention, SD = Standard deviation, CON = Control, SMD = Standardised mean difference, SE = Standard error, IV = Inverse variance, CI = Confidence interval

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Figure 3b | VO2max response comparison between END versus HIIT and END versus RET

Author	Date	Study Design Type	Intervention Duration	Intervention	Indicator	END			HIIT			Derivative Statistics				95% CI	
						Mean	SD	Sample Size	Mean	SD	Sample Size	Pooled SD	SMD	SE	IV	Lower	Upper
Berger	2005	C	6 weeks	END vs. HIIT	VO2max	40.3	5.3	8	42.5	8.9	8	7.32	-0.30	1.83	-0.30	-3.89	3.29
Bonafiglia	2016	RCT	3 weeks x 2	END vs. HIIT	VO2max	43.9	6.4	21	42.9	5.5	21	5.97	0.17	0.92	0.17	-1.63	1.97
Burgomaster *	2008	C	6 weeks	END vs. HIIT	VO2max	45	2	10	44	2	10	2.00	0.50	0.45	0.50	-0.38	1.38
Dunham	2012	RCT	4 weeks	END vs. HIIT	VO2max	34.3	6.7	7	36.5	6.7	8	6.70	-0.33	1.73	-0.33	-3.72	3.06
MacPherson	2010	C	6 weeks	END vs. HIIT	VO2max	49.5	8.3	10	52.2	6.5	10	7.45	-0.36	1.67	-0.36	-3.63	2.91
Matsuo	2013	RCT	8 weeks	END vs. HIIT	VO2max	41.9	5.6	14	42	6.8	14	6.23	-0.02	1.18	-0.02	-2.33	2.29
McKay	2009	RCT	3 weeks	END vs. HIIT	VO2max	46	6	6	48	4	6	5.10	-0.39	1.47	-0.39	-3.27	2.49
Nybo	2010	C	12 weeks	END vs. HIIT	VO2max	42.2	1.8	9	41.4	2.2	8	2.01	0.40	0.49	0.40	-0.56	1.36
Sandvel	2012	RCT	8 weeks	END vs. HIIT	VO2max	49.7	1.8	12	53.5	1.7	11	1.75	-2.17	0.37	-2.17	-2.9	-1.44
Shepherd	2013	RCT	6 weeks	END vs. HIIT	VO2max	48.2	5	8	45.1	2.3	8	3.89	0.80	0.97	0.80	-1.1	2.7
Trapp	2008	RCT	15 weeks	END vs. HIIT	VO2max	36.9	1.5	15	36.4	1.5	15	1.50	0.33	0.27	0.33	-0.2	0.86
Warburton **	2004	RCT	12 weeks	END vs. HIIT	VO2max	49.6	5.8	6	47.3	13	6	9.87	0.23	2.85	0.23	-5.36	5.82
Nybo	2010	C	12 weeks	END vs. RET	VO2max	42.2	1.8	9	37.9	3.3	8	2.66	1.62	0.64	1.62	0.37	2.87
Hautala	2006	RCT (crossover)	2 weeks	END vs. RET	VO2max	37	7	73	36	7	73	7.00	0.14	0.58	0.14	-1	1.28

Legend

Notes

** 6 week values obtained. ** Total (male and female) values incorporated.*

Abbreviations

C = Case-control, RCT = Randomised controlled trial, END = Endurance, HIIT = High-intensity interval training, RET = Resistance training, SD = Standard deviation, SMD = Standardised mean difference, SE = Standard error, IV = Inverse variance, CI = Confidence Interval

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Figure 3c | Per-exercise modality comparison in response with alternative response indicators

Author	Date	Study Design Type	Intervention Duration	Intervention	Indicator	END			HIIT			Derivative Statistics				95% CI	
						Mean	SD	Sample Size	Mean	SD	Sample Size	Pooled SD	SMD	SE	IV	Lower	Upper
Kohrt *	1991	C	36-52 weeks	END	HRrest	76.1	12.0	110	78.4	###	81	12.28	-0.18	0.89	-0.18	-1.92	1.56
Nybo	2010	C	12 weeks	END	HRrest	53	2	9	61	3	11	2.55	-3.14	0.57	-3.14	-4.26	-2.02
Kohrt *	1991	C	36-52 weeks	END	HRmax	163.4	12.4	110	163.9	###	81	11.73	-0.05	0.85	-0.05	-1.72	1.62
Nybo **	2010	C	12 weeks	END	HRmax												
Lo †	2011	RCT	24 weeks	END	HRmax	195.1	7.5	10	192.1	10	10	8.90	0.34	1.99	0.34	-3.56	4.24
Warburton	2004	RCT	12 weeks	END	HRmax	188	12	6	185	6	8	9.49	0.32	2.54	0.32	-4.66	5.3
Berger	2005	C	6 weeks	END	HRmax	170	11	8	177	12	7	11.51	-0.61	2.97	-0.61	-6.43	5.21
Hautala	2006	RCT (crossover)	Two weeks	END	HRmax	181	9	73	180	11	18	10.05	0.10	1.05	0.10	-1.96	2.16
Berger	2005	C	6 weeks	END	LT	5.2	1.1	8	4.6	0.6	7	0.89	0.68	0.23	0.68	0.23	1.13
Warburton	2004	RCT	12 weeks	END	PO	291	71	6	229	39	8	57.28	1.08	####	1.08	-28.93	31.09
Berger	2005	C	6 weeks	END	PO	194	41	8	174	50	7	45.72	0.44	####	0.44	-22.71	23.59
Nybo	2010	C	12 weeks	HIIT	HRrest	52	2	8	61	3	11	2.55	-3.53	0.58	-3.53	-4.67	-2.39
Weatherwax (Standardised)	2018	RCT	12 weeks	HIIT	HRrest	68.2	8	20	69.5	7.5	8	7.75	-0.17	1.47	-0.17	-3.05	2.71
Weatherwax (Individualised)	2018	RCT	12 weeks	HIIT	HRrest	68.1	11.4	19	69.5	7.5	8	9.65	-0.15	1.86	-0.15	-3.8	3.5
Nybo **	2010	C	12 weeks	HIIT	HRmax												
Berger	2005	C	6 weeks	HIIT	HRmax	178	14	8	177	12	7	13.04	0.08	3.37	0.08	-6.53	6.69
Astorino ‡	2018	C	3 weeks	HIIT	HRmax	182.5	11.8	14	182.7	7.2	14	9.77	-0.02	1.85	-0.02	-3.65	3.61

Legend

Notes

** Pooled mean and SD calculated between male and female groups from "Final" datasets. Reported to 1 d.p. † Reported to 1 d.p.*

*** Omitted - measured submaximal HR at exertion with 9.5kmph training velocity. ‡ Derived from RAMP (incremental exercise test) data.*

Abbreviations

C = Case-control, RCT = Randomised controlled trial, END = Endurance, HIIT = High-intensity interval training, RET = Resistance training,
HRrest = Heart rate at rest, Hrmax = Maximal heart rate, LT = Lactate threshould , PO = Power output , SD = Standard deviation,
SMD = Standardised mean difference, SE = Standard error, IV = Inverse variance, CI = Confidence interval

continued

Figure 3c | Per-exercise modality comparison in response with alternative response indicators (cont.)

Author	Date	Study Design Type	Intervention Duration	Intervention	Indicator	END			HIIT			Derivative Statistics				95% CI	
						Mean	SD	Sample Size	Mean	SD	Sample Size	Pooled SD	SMD	SE	IV	Lower	Upper
Weatherwax (Standardised)	2018	RCT	12 weeks	HIIT	HRmax	164.9	15.1	20	170.1	11	8	13.21	-0.39	2.50	-0.39	-5.29	4.51
Weatherwax (Individualised)	2018	RCT	12 weeks	HIIT	HRmax	169.2	14.4	19	170.1	11	8	12.81	-0.07	2.47	-0.07	-4.91	4.77
Berger	2005	C	6 weeks	HIIT	LT	5.1	0.8	8	4.6	0.6	7	0.71	0.71	0.18	0.71	0.36	1.06
Ziemann	2011	RCT	6 weeks	HIIT	PO	307	29	10	250	31	11	30.02	1.90	6.55	1.90	-10.94	14.74
Warburton	2004	RCT	12 weeks	HIIT	PO	305	67	6	229	39	8	54.82	1.39	####	1.39	-27.32	30.1
Berger	2005	C	6 weeks	HIIT	PO	224	73	8	174	**50**	**7**	62.57	0.80	####	0.80	-30.85	32.45
Astorino ‡	2018	C	3 weeks	HIIT	PO	267	57	14	287	60	14	58.52	-0.34	####	-0.34	-22.02	21.34
Nybo	2010	C	12 weeks	RET	HRrest	56	2	8	61	3	11	2.55	-1.96	0.58	-1.96	-3.1	-0.82
Nybo **	2010	C	12 weeks	RET	HRmax												
Lo †	2011	RCT	24 weeks	RET	HRmax	196.2	9.2	10	192.1	10	10	9.66	0.42	2.16	0.42	-3.81	4.65
Hautala	2006	C	2 weeks	RET	HRmax	180	10	73	180	11	18	10.51	0.00	1.10	0.00	-2.16	2.16

Legend

Notes

** Pooled mean and SD calculated between male and female groups from "Final" datasets. Reported to 1 d.p. † Reported to 1 d.p.*

*** Omitted - measured submaximal HR at exertion with 9.5kmph training velocity. ‡ Derived from RAMP (incremental exercise test) data.*

Abbreviations

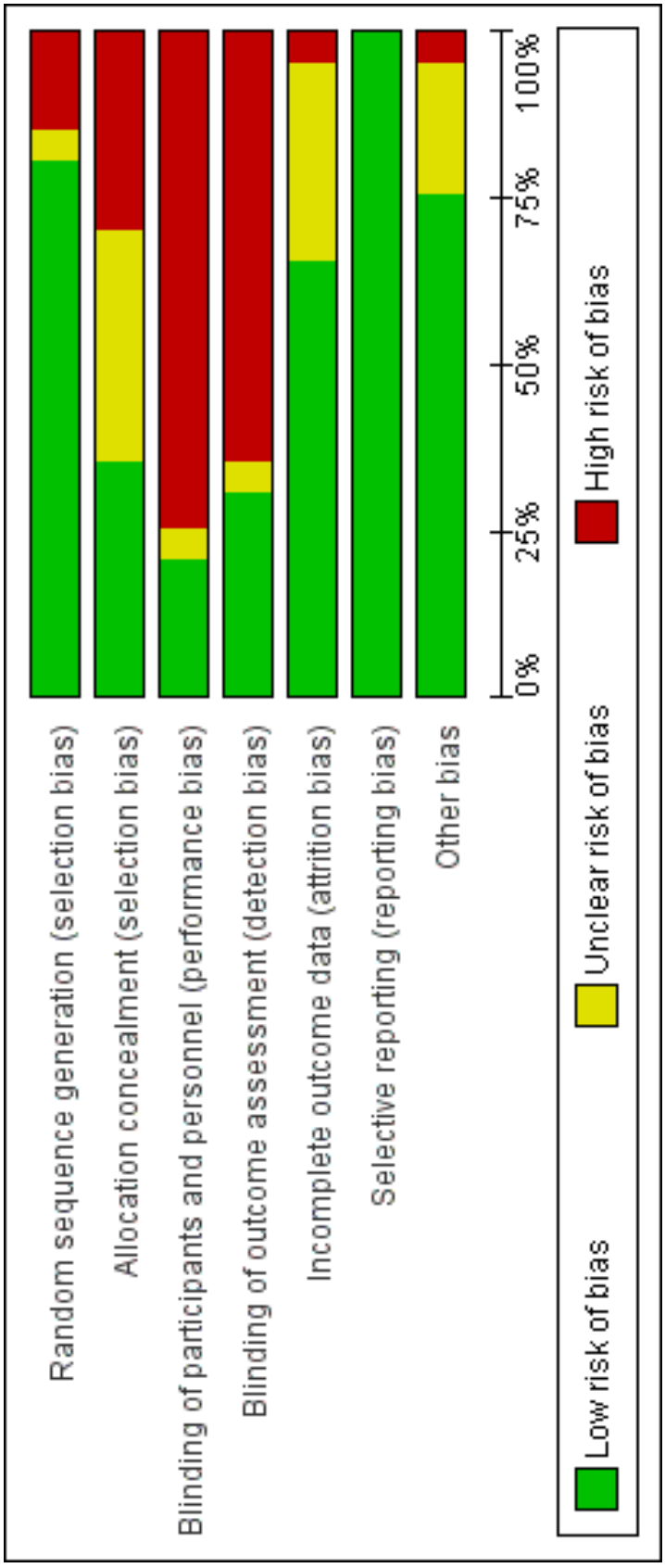
C = Case-control, RCT = Randomised controlled trial, END = Endurance, HIIT = High-intensity interval training, RET = Resistance training, HRrest = Heart rate at rest, Hrmax = Maximal heart rate, LT = Lactate threshould , PO = Power output , SD = Standard deviation, SMD = Standardised mean difference, SE = Standard error, IV = Inverse variance, CI = Confidence interval

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Supplementary 4 – GRADE score results for all studies.

Lead Author	Publication Date	Bias Domain				
		Risk of Bias	Imprecision	Inconsistency	Indirectness	Publication Bias
Randomised Control Trials						
Sisson	2009	⊕⊕⊕⊕	no CI reported	⊕⊕⊕	⊕⊕⊕⊕	N/A*
Earnest	2011	⊕⊕⊕⊕	⊕⊕⊕	⊕⊕⊕	⊕⊕⊕	N/A*
Pandey	2015	⊕⊕	no CI reported	⊕⊕⊕	⊕⊕⊕	N/A*
Montero	2017	⊕⊕	⊕	⊕⊕⊕⊕	⊕⊕⊕	N/A*
Bonafiglia	2016	⊕⊕	no CI reported	⊕⊕⊕⊕	⊕⊕⊕	N/A*
Ross	2015	⊕⊕⊕⊕	no CI reported	⊕⊕⊕	⊕⊕⊕⊕	N/A*
Gurd	2016	⊕⊕⊕⊕	no CI reported	⊕⊕⊕	⊕⊕⊕	N/A*
Osei-Tutu & Campagna	2004	⊕⊕⊕	no CI reported	⊕⊕⊕	⊕⊕	N/A*
Trapp	2008	⊕⊕⊕	no CI reported	⊕⊕⊕⊕	⊕⊕	N/A*
Metcalfe	2012	⊕⊕⊕	no CI reported	⊕⊕⊕	⊕⊕	N/A*
Ziemann	2011	⊕⊕⊕	no CI reported	⊕⊕⊕	⊕	N/A*
Lo	2011	⊕⊕⊕	no CI reported	⊕⊕⊕	⊕⊕	N/A*
McKay	2009	⊕⊕⊕	no CI reported	⊕⊕⊕⊕	⊕⊕	N/A*
Dunham	2012	⊕⊕⊕	no CI reported	⊕⊕	⊕⊕⊕	N/A*
Shepherd	2013	⊕⊕	no CI reported	⊕⊕	⊕⊕	N/A*
Warburton	2004	⊕⊕	no CI reported	⊕⊕⊕	⊕⊕	N/A*
Matsuo	2013	⊕⊕	no CI reported	⊕⊕	⊕⊕	N/A*
O'Donovan	2005	⊕⊕⊕	no CI reported	⊕⊕⊕	⊕⊕	N/A*
Sandvei	2012	⊕⊕⊕	no CI reported	⊕⊕	⊕⊕	N/A*
Weatherwax	2018	⊕⊕	no CI reported	⊕⊕	⊕⊕⊕	N/A*
Cohort or Case Control						
Scharhag-Rosenberger	2012	⊕⊕⊕	no CI reported	⊕⊕⊕⊕	⊕⊕⊕	N/A*
Higgins	2015	⊕⊕⊕	no CI reported	⊕⊕⊕	⊕⊕⊕	N/A*
Astorino	2018	⊕⊕⊕	no CI reported	⊕⊕⊕⊕	⊕⊕⊕	N/A*
Hautala	2006	⊕⊕⊕	no CI reported	⊕⊕	⊕⊕	N/A*
Nybo	2010	⊕⊕	no CI reported	⊕⊕⊕	⊕⊕⊕	N/A*
Burgomaster	2008	⊕⊕⊕	no CI reported	⊕⊕⊕	⊕⊕	N/A*
Macpherson	2010	⊕⊕⊕	no CI reported	⊕⊕⊕	⊕⊕⊕	N/A*
Berger	2006	⊕⊕⊕	no CI reported	⊕⊕⊕⊕	⊕⊕⊕	N/A*
Yan	2017	⊕⊕⊕	no CI reported	⊕⊕⊕	⊕⊕⊕	N/A*

Legend		
Notes	High	⊕⊕⊕⊕
*N/A * - Not a systematic review.*	Moderate	⊕⊕⊕
Abbreviations	Low	⊕⊕
N/A - Not applicable, CI - Confidence Interval.	Very Low	⊕



Supplementary 5 | Risk of Bias category results for all studies

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	6-8
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	8
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	9
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	9
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	9
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	9
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	10
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	10
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	10
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	11
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	11
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	11

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	N/A
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	13
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	13
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	13
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	13
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	14
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	N/A
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	16-19
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	19
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	19-20

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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